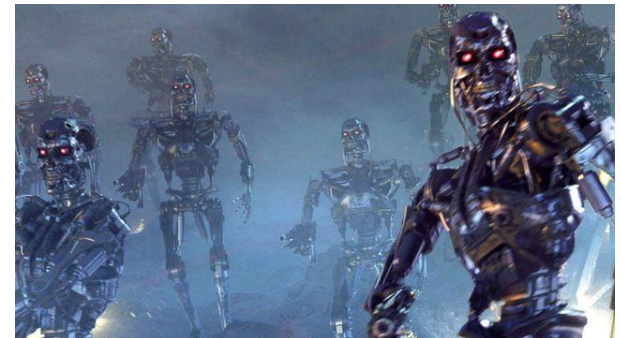
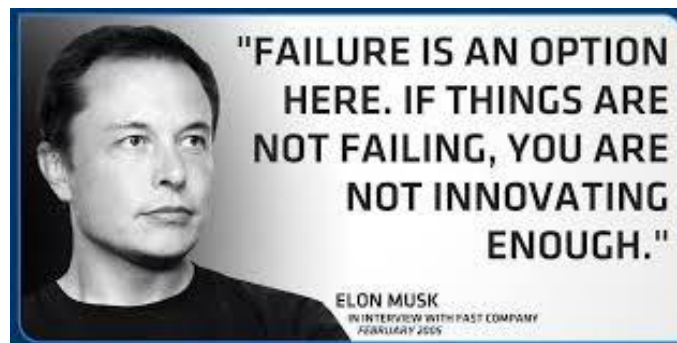




Deep Learning: The end of the « Black Box Era »

28/08/2018 //

Olivier Croissant





1. Introduction

- Universal properties of NN

2. Information Analysis

- Mutual Information
- Information Bottleneck

3. Wavelets Approach

- Closure Formula

4. Renormalization Application

- Wilson renormalization group
- Ising model – Deep Belief nets
- Power laws



We desperately need a Theory

Why DNN work so well?

How can they be improved?

- Design principle

- I. What(determnes the number & width of the layers ?
- II. What determines the connectivity and inter-layer connections ?

- Interpretability

- I. What do the layers/neurons capture/represent ?
- II. How can we monitor the learning of the network, and decide when it reached its optimum ?

- Better algorithms

- I. Is stochastic gradient descent the best algorithm we can do ?
- II. How can we determine the best minibatch size and the best dropout percentage ?



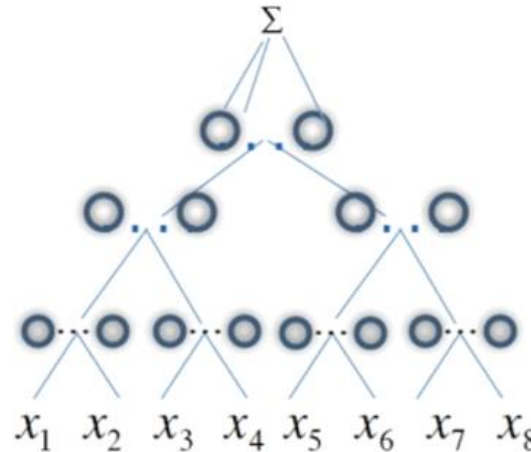
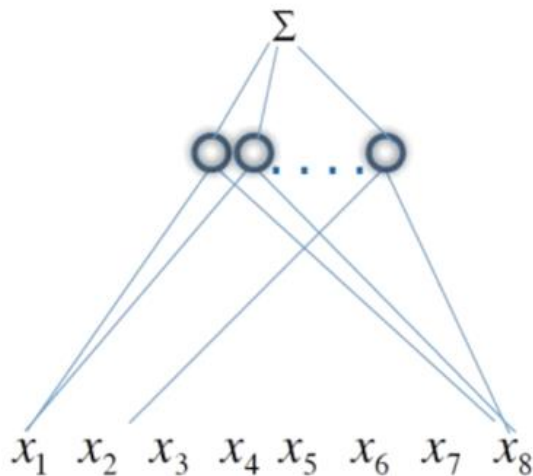
Stochastic dynamics of the training process

- Convergence of SGD to a gibbs (max entropy)
- The compression of information during the learning.
 - Input Compression Bounds
 - PAC-like Generalization Bounds
 - What is a typical input
- The benefits of hidden layers
 - the role of diffusion of the noise at every level
- Many local optimums are good enough..
 - Global optimum is not needed
 - The convergence go through many stages..

Universal properties of NN



Theorem *Shallow, one-hidden layer networks with a nonlinear $\phi(x)$ which is not a polynomial are universal. Arbitrarily deep networks with a nonlinear $\phi(x)$ (including polynomials) are universal.*



$$\phi(x) = \sum_{i=1}^r c_i | \langle w_i, x \rangle + b_i |_+$$

Curse of Dimensionality



$$f(x_1, x_2, \dots, x_8)$$

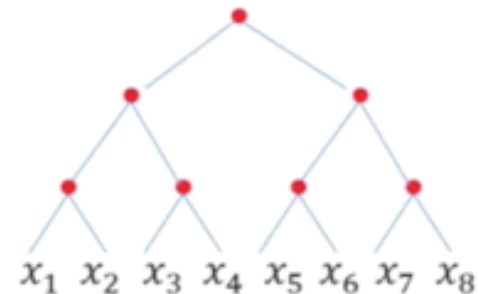
Curse of dimensionality

Both shallow and deep network can approximate a function of d variables equally well. The number of parameters in both cases depends exponentially on d as $O(\varepsilon^{-d})$.



Hierarchically local compositionality (HLC)

$$f(x_1, x_2, \dots, x_8) = g_3(g_{21}(g_{11}(x_1, x_2), g_{12}(x_3, x_4))g_{22}(g_{11}(x_5, x_6), g_{12}(x_7, x_8)))$$



Theorem (informal statement)

Suppose that a function of d variables is hierarchically, locally, compositional. Both shallow and deep network can approximate f equally well. The number of parameters of the shallow network depends exponentially on d as $O(\epsilon^{-d})$ with the dimension whereas for the deep network dance is $O(d\epsilon^{-2})$

Information Theory



Kullback-Leibler Divergence



$$D_{KL}[p||q] = \int_{-\infty}^{\infty} p(x) \text{Log} \frac{p(x)}{q(x)} dx$$

$$D_{KL}[p||p] \geq 0 \quad \text{et} \quad D_{KL}(p||p) = 0$$

But it is not a distance but triangular inégalité does not work

Is not symmetric

let $q(x)$ a prob. distribution defined on E et $F \subset E$

Let $q_F(x)$ a prob. distribution restricted to F (conditionned by F)

$$\text{Then } D_{KL}[q||q_F] = \text{Log} \frac{1}{\text{Prob}[F]}$$

- This is the supplementary information: conditionned by F (in bits)
- This is the Renyi divergence for $\alpha = 1$, $D_\alpha[p||q] = \frac{1}{1-\alpha} \int_{-\infty}^{\infty} p(x)^\alpha q(x)^{1-\alpha} dx$ Hellinger

The topology generated by D_{KL} and D_α are not the same : $D_{1/2} \rightarrow \text{Distance}$ $\downarrow$

$$\lim_{\theta_1 \rightarrow \theta} \frac{1}{(\theta_1 - \theta)^2} D_\alpha[p_{\theta_1}||p_\theta] = \frac{\alpha}{2} J(\theta) \leftarrow \text{Fisher Information}$$

KL-Divergence and Crossed Entropy..



Choosing
The right tool



Claude Shannon

$$D_{KL}[p||q] = \int p(x) \log p(x) dx - \int p(x) \log q(x) dx$$

Measure a spread

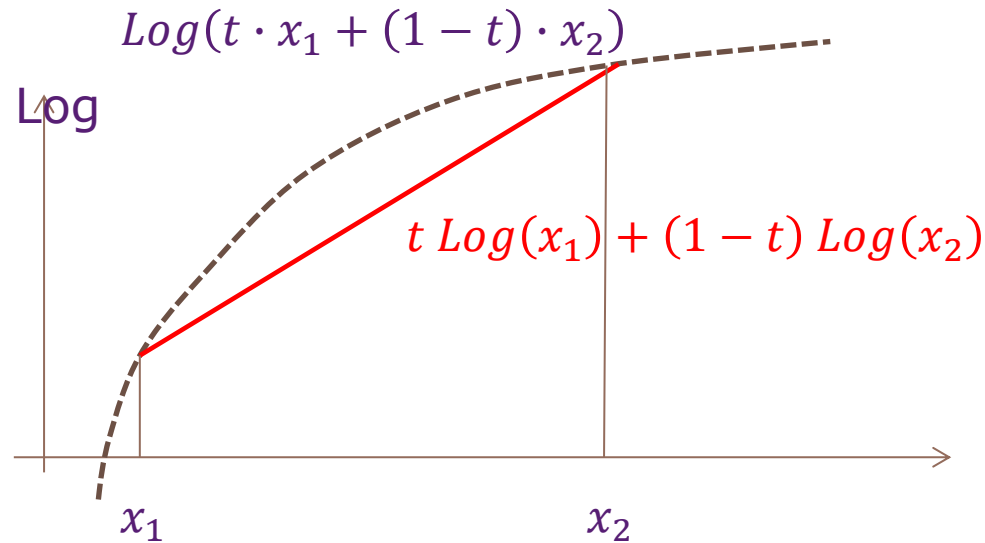
$$D_{KL}[p||q] = H(p) - H(p, q)$$

Entropy

Crossed Entropy

- An Efficient criteria
- Classification problems
- Far Gradient still active

Jensen Inequality



2 points

N points > 2

Jensen :

$$F \text{ concave} \Rightarrow f(\mathbb{E}[X]) \geq \mathbb{E}[f(X)]$$

Inférence variationnelle (mécanique)



$$\text{Log } p(x) - D_{KL}[q(z)||p(z|x)] = (\mathbb{E}_q[\text{Log } p(z,x)] - \mathbb{E}_q[\text{Log } q(z)])$$

Or $p(z, x) = p(x|z) p(z)$

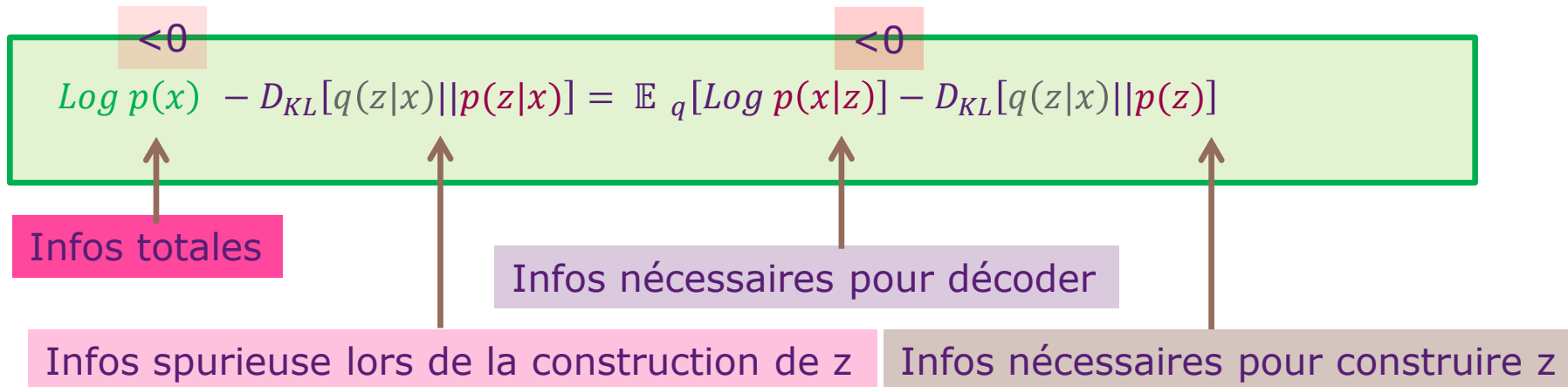
$$\text{Log } p(x) - D_{KL}[q(z)||p(z|x)] = (\mathbb{E}_q[\text{Log } p(x|z) + \text{Log } p(z)] - \mathbb{E}_q[\text{Log } q(z)])$$

Or $D_{KL}[q(z)||p(z)] = \mathbb{E}_q[\text{Log } p(z)] - \mathbb{E}_q[\text{Log } q(z)]$

$$\text{Log } p(x) - D_{KL}[q(z)||p(z|x)] = \mathbb{E}_q[\text{Log } p(x|z)] - D_{KL}[q(z)||p(z)]$$



On peut faire dépendre q de x



Variational Inference (Boundary)



X observed
Z hidden

Z is not observed

$$\text{Log } p(x) = \text{Log} \left[\int_z p(x, z) dz \right]$$

q(z) arbitrary

$$= \text{Log} \left[\int_z p(x, z) \frac{q(z)}{q(z)} dz \right]$$

$$= \text{Log} \left[\mathbb{E}_q \left[\frac{p(x, z)}{q(z)} \right] \right]$$

Entropy of z

Jensen

$$\geq \mathbb{E}_q [\text{Log}[p(x, z)]] - \underbrace{\mathbb{E}_q [\text{Log}[q(z)]]}_{\text{Entropy of z}} \equiv \text{ELBO}(p, q)$$

Then maximising expectation of the low bound gives a low bound on the likelihood which is a good approximation

Relationship with KL divergence



We want to approximate $p(z|x) = \frac{p(z,x)}{p(x)}$ by $q(z)$

Definition

$$\begin{aligned} D_{KL}[q(z)||p(z|x)] &\stackrel{\text{Definition}}{=} \mathbb{E}_q \left[\text{Log} \frac{q(z)}{p(z|x)} \right] \\ &= \mathbb{E}_q [\text{Log} q(z)] - \mathbb{E}_q [\text{Log} p(z|x)] \\ &= \mathbb{E}_q [\text{Log} q(z)] - \mathbb{E}_q [\text{Log} p(z,x)] + \text{Log} p(x) \end{aligned}$$

Does't depend on q

$$\text{Equation K} \quad = -(\mathbb{E}_q [\text{Log} p(z,x)] - \mathbb{E}_q [\text{Log} q(z)]) + \text{Log} p(x)$$

Only varying $q(z)$
Then Minimizing ELBO $\Leftrightarrow$ Maximizing KL divergence

Mutual Information Properties



Definition

$$I(X; Y) = D_{KL}[p(x, y) || p(x)p(y)]$$

The information improvement compared to an independent case

Data processing Inequality (DPI)

For any markov chains $X \rightarrow Y \rightarrow Z$

$$I(X; Y) \geq I(X; Z)$$

There is less information on X in Z than in Y

Invariance by reparametrisation

$$I(X; Y) = I(\varphi(X); \psi(Y)) \text{ For any invertible } \varphi \text{ and } \psi$$

How do we measure mutual information

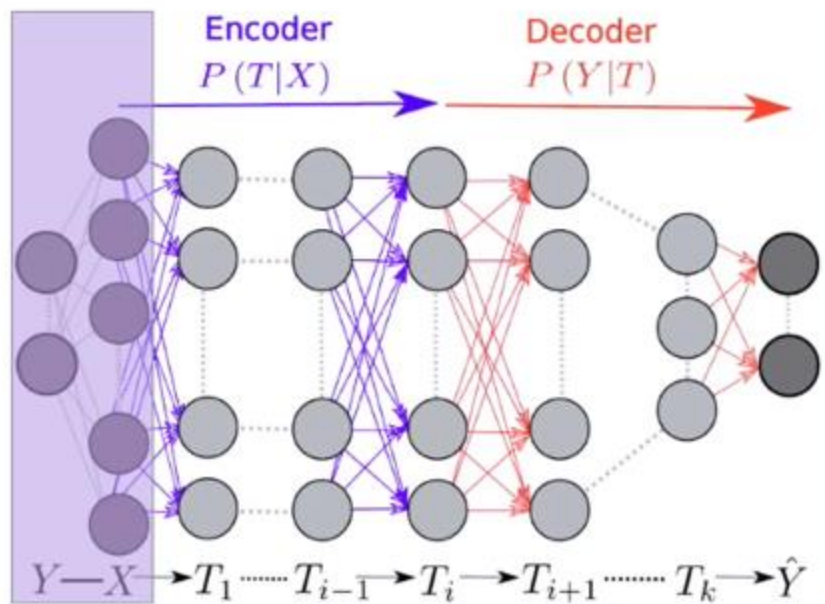


- The representation invariance of the mutual information raises an interesting question.
- Obviously, the computational complexity of learning is not representation invariant (think about learning from encrypted patterns). **Thus, Information measures can't tell the whole story.**
- Our experiments crucially depends on how we estimate information. We consider 3 types of estimations: **(1) binning the variables. (2) adding noise / stochasticity (3) parametric approximations.** In our experiments we quantized/bin the neuronal output values.
- All assume compressibility/refineability of the variables. They are not robust to arbitrary invertible transformations!
- The assertion that the layers are invertible transformations of the input is NOT robust to small noise and misleading. Binning or assuming stochastic mapping is essential for our information theoretic approach.
- Moreover, the IB is trivial (uninteresting) for completely deterministic rules! I argue that our theory predicts that *without additional structural information on the patterns*, DL can't work for completely deterministic rules, as they can't be distinguished from random (fully mixing) rules!

Introduction of Information Plane



Each layer is characterized by its Encoder & Decoder Information



Theorem (Information Plane):

For large typical X , the sample complexity of a DNN is completely determined by the encoder mutual information, $I(X;T)$, of the last hidden layer; the accuracy (generalization error) is determined by the decoder information, $I(T;Y)$, of the last hidden layer.

The complexity of the problem shifts from the decoder to the encoder, across the layers...

Rethinking Learning Theory



“Old” Generalization bounds:

$$\varepsilon^2 < \frac{\log |H_\varepsilon| + \log 1/\delta}{2m}$$

ε - generalization error

δ - confidence

m - number of training examples

H_ε - ε -cover of the Hypothesis class

typically we assume: $|H_\varepsilon| \sim \left(\frac{1}{\varepsilon}\right)^d$

d - the class (VC,...) dimension

... Don't work for Deep Learning!

Higher expressivity - worse bound!



New: Input Compression bound:

$$|H_\varepsilon| \sim 2^{|X|} \rightarrow 2^{|T_\varepsilon|}$$

T_ε - ε -partition of the input variable X

Information Theory: $|T_\varepsilon| \sim 2^{I(T_\varepsilon; X)}$

$$\varepsilon^2 < \frac{2^{I(T_\varepsilon; X)} + \log 1/\delta}{2m}$$

... K bits of compression of X are like
a factor of 2^K training examples!

Rethinking Learning Theory



What are “large typical” patterns?

Typicality emerges when the underlying pattern distribution can be asymptotically expressed as a long product of localized conditional probabilities.

E.g. Markov Random Fields, Hidden Markov Models, *pairwise* interaction Hamiltonians in physics, all common Graphical models, etc.

In our case it includes images, speech & text, long molecular sequences, signals generated by localized dynamic systems, etc.

Then, the Shannon-McMillen limit for the entropy exists:

$$\lim_{n \rightarrow \infty} -\frac{1}{n} \log p(x_1, \dots, x_n) = H(X)$$

and *almost all* patterns are *typical* with probability:

$$p(x_1, \dots, x_n) \approx 2^{-nH(X)}$$

and also for large enough typical partitions, T :

$$p(x_1, \dots, x_n | T) \approx 2^{-nH(X|T)}$$

Information Bottleneck (IB) Method



(1) Approximate Minimal Sufficient Statistics:

Markov chain: $Y \rightarrow X \rightarrow S(X) \rightarrow \hat{X}$

$$\hat{X} = \arg \min_{S(X) : I(S(X); Y) = I(X; Y)} I(S(X); X)$$

Relaxation - given $p(X, Y)$:

$$\hat{X} = \arg \min_{p(\hat{x}|x)} I(\hat{X}; X) - \beta I(\hat{X}; Y), \quad \beta > 0$$

(Shamir, Sabato, T., TCS 2010)

(2) A Rate-Distortion problem with KL- divergence distortion:

$$d_{IB}(x, \hat{x}) = D[p(y|x) \| p(y|\hat{x})]$$

(Bachrach, Navot, T., COLT 2006)

(3) The ONLY distributional quantization measure which satisfy

both DPI (f-divergences) and Statistical Consistency (Bregman divergences)

(Harremoës-T., ISIT 2008)



The Information Bottleneck Optimality Bound

The IB bound optimality equations:

$$\min_{p(\hat{x}|x); Y \rightarrow X \rightarrow \hat{X}} I(\hat{X}; X) - \beta I(\hat{X}; Y), \quad \beta > 0$$

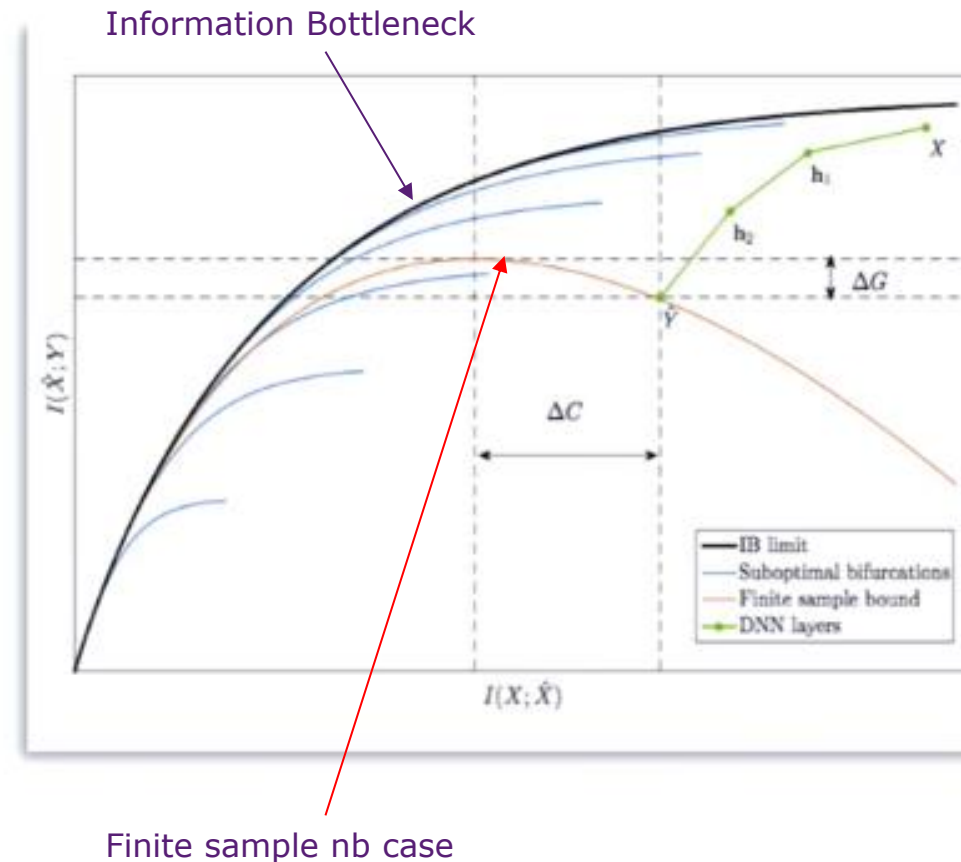
$$p(x|\hat{x}) = \frac{p(x)}{Z(x, \beta)} \exp(-\beta D[p(y|x) \| p(y|\hat{x})])$$

$$Z(x, \beta) = \sum_{\hat{x}} p(\hat{x}) \exp(-\beta D[p(y|x) \| p(y|\hat{x})])$$

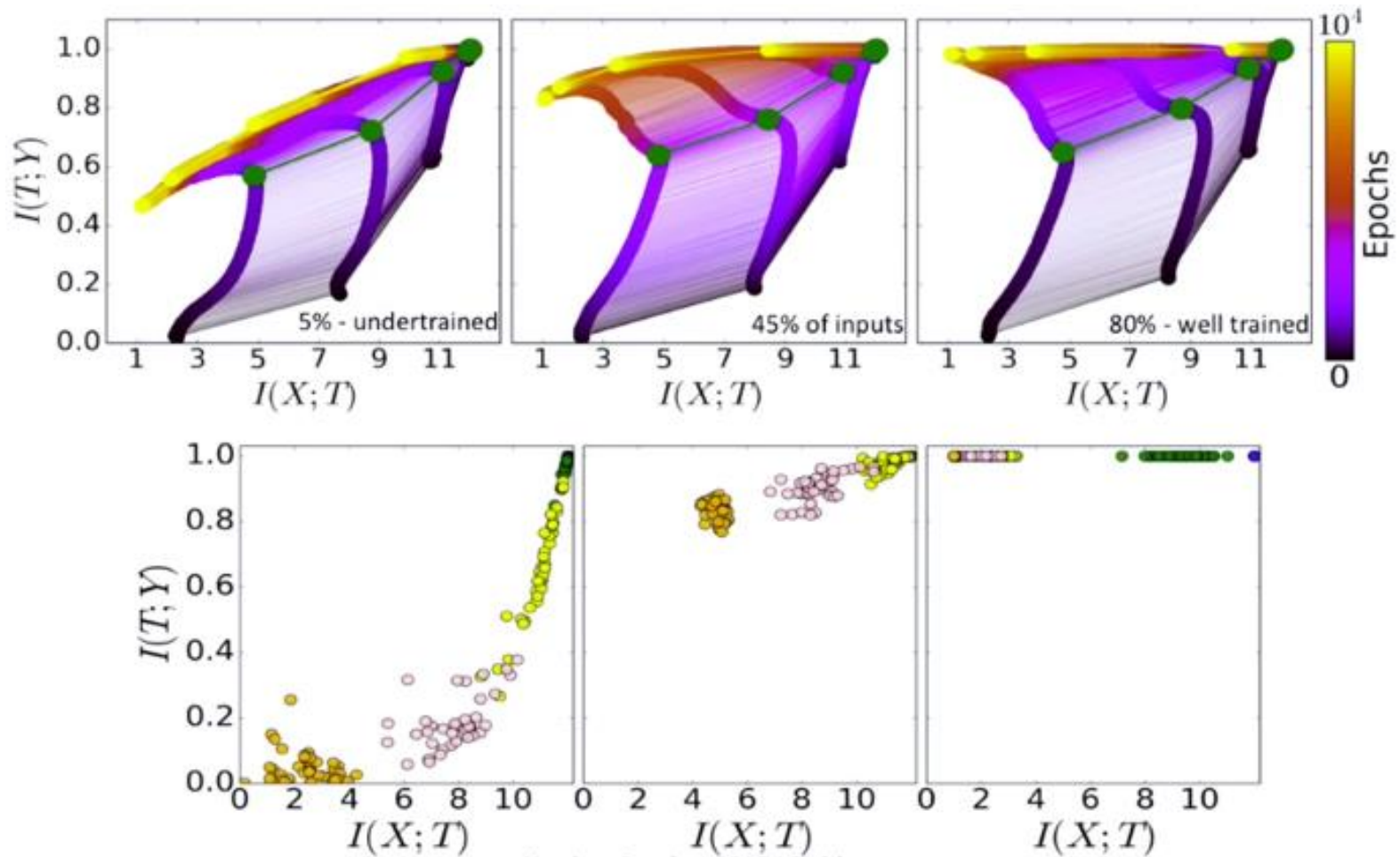
$$p(\hat{x}) = \sum_x p(\hat{x}|x) p(x)$$

$$p(y|\hat{x}) = \sum_x p(y|x) p(x|\hat{x})$$

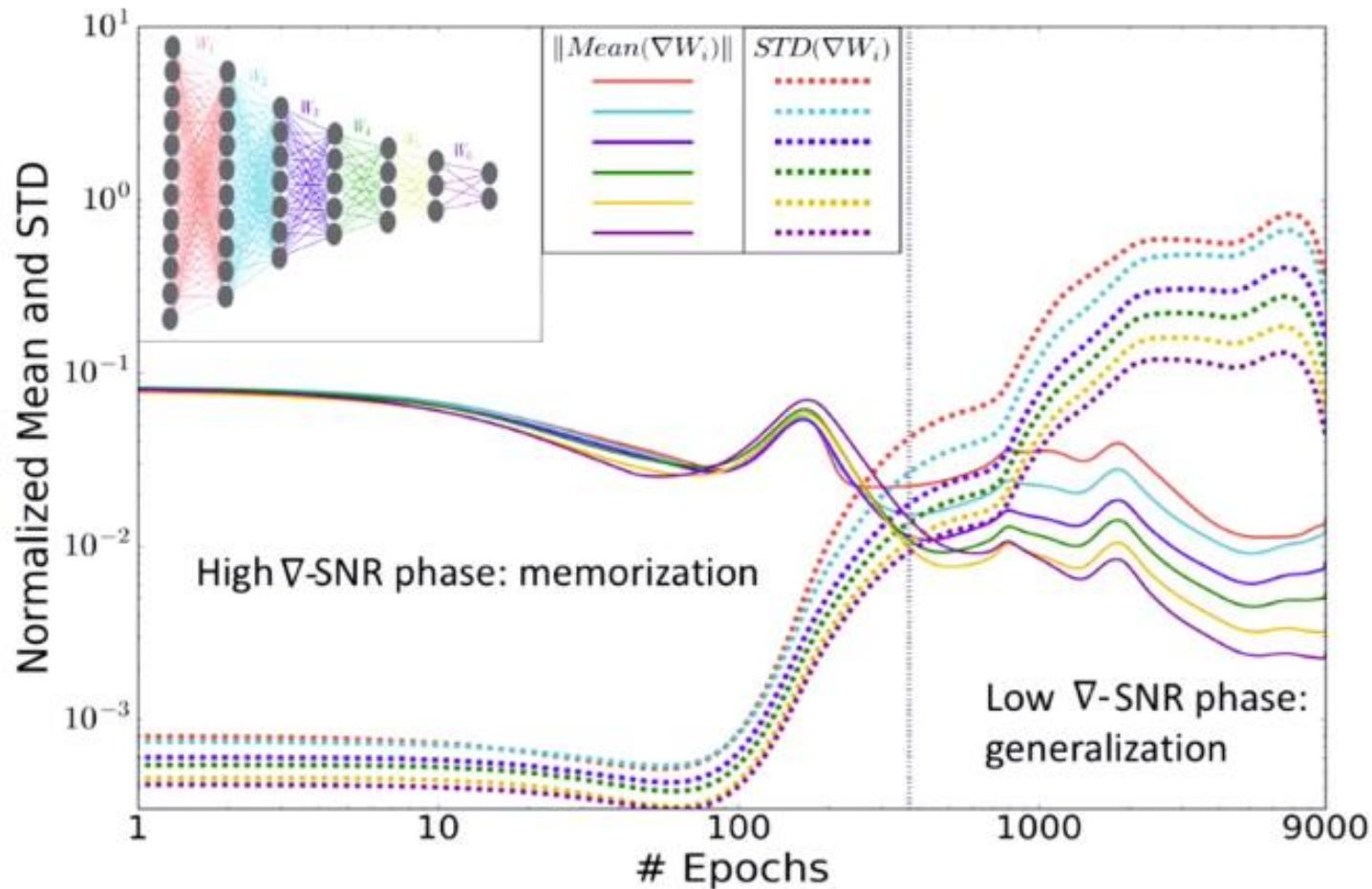
Solved by Arimoto-Blahut like iterations,
but with possibly sub-optimal solutions, bifurcations (!),



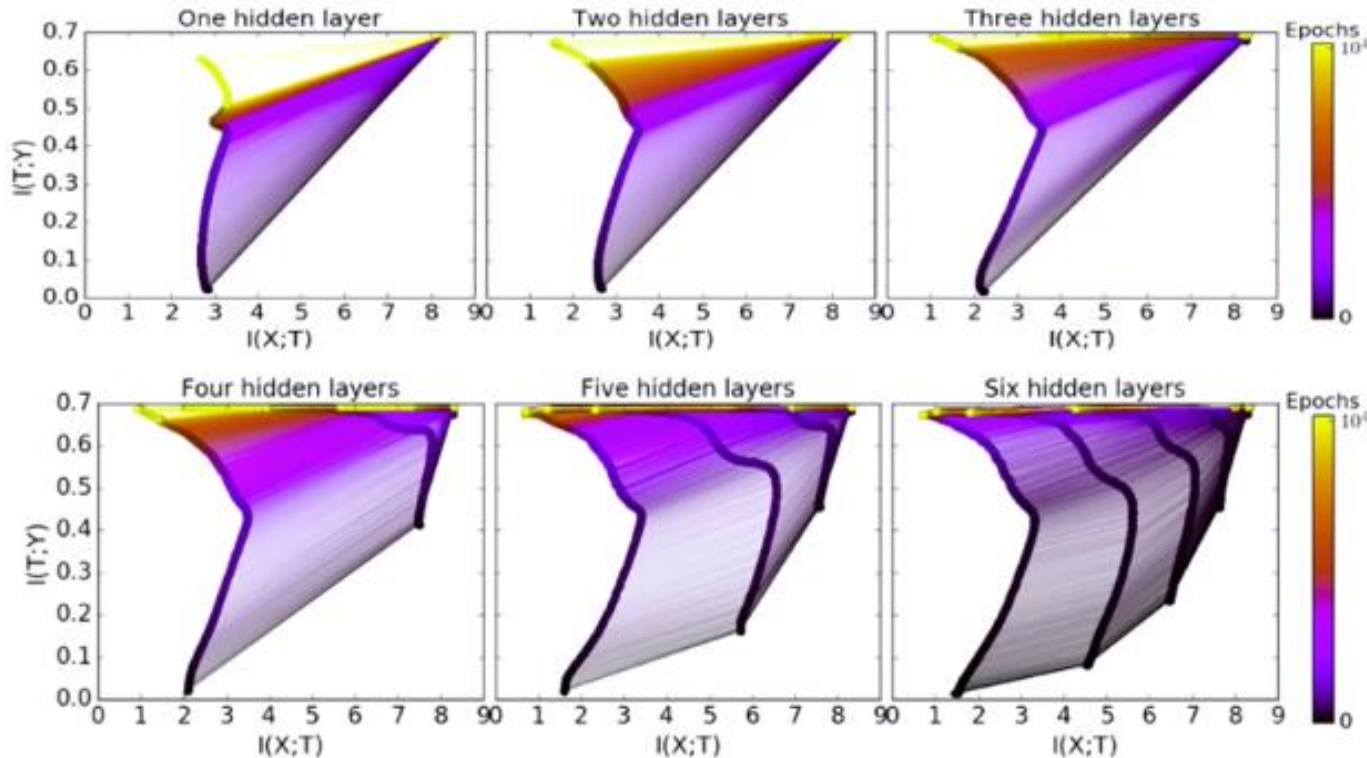
Information Plane in Action



Appearance of the generalization phase



Importance of Hidden layers



More layers take much FEWER training epochs for good generalization.

The optimization time depend super-linearly (exponentially?) on the compressed information, ΔI_X , for each layer.

Relaxation times and Hidden layers



Noisy relaxation (SGD): $\frac{\partial W_k}{\partial t} = -\nabla E(W_k) + \beta_k^{-1} \xi(t)$, layer k , $\xi \sim N(0,1)$

$\Rightarrow$ Maximum Entropy (via Focker-Planck): $P_{Gibbs}(W_k) \propto \exp(-\beta_k E(W_k))$,

Relaxation time for non-strongly convex error: $\Delta t_k \sim \exp(\Delta S_k)$

Denote the layer compression be: $\Delta S_k = I(X; T_k) - I(X; T_{k-1})$

Since $\exp\left(\sum_k \Delta S_k\right) \gg \sum_k \exp(\Delta S_k) > \max_k \exp(\Delta S_k) \Rightarrow$

Exponential boost in the relaxation time with K layers!

Compression Through Stochastic Relaxation



Noisy relaxation (SGD) on training error (with m examples):

$$\frac{\partial W_k}{\partial t} = -\nabla E(W_k | X^{(m)}) + \beta_k^{-1} \xi(t), \quad \text{layer } k, \quad \xi \sim N(0,1), \quad \beta_k - \text{decoder } k\text{-layer noise}$$

$$\Rightarrow \text{Maximum Entropy: } P_{\text{Gibbs}}(W_k | X^{(m)}) \propto \exp(-\beta_k E(W_k | X^{(m)}))$$

with additive [cross-entropy] training error and i.i.d. samples, using Bayes rule with the quenched W :

$$P_{\text{Gibbs}}(X | W_k) = P_{\text{Gibbs}}(X | T_k) \propto \exp(-\tilde{\beta}_k D_{KL}[p(Y | X) \| p(Y | T)])$$

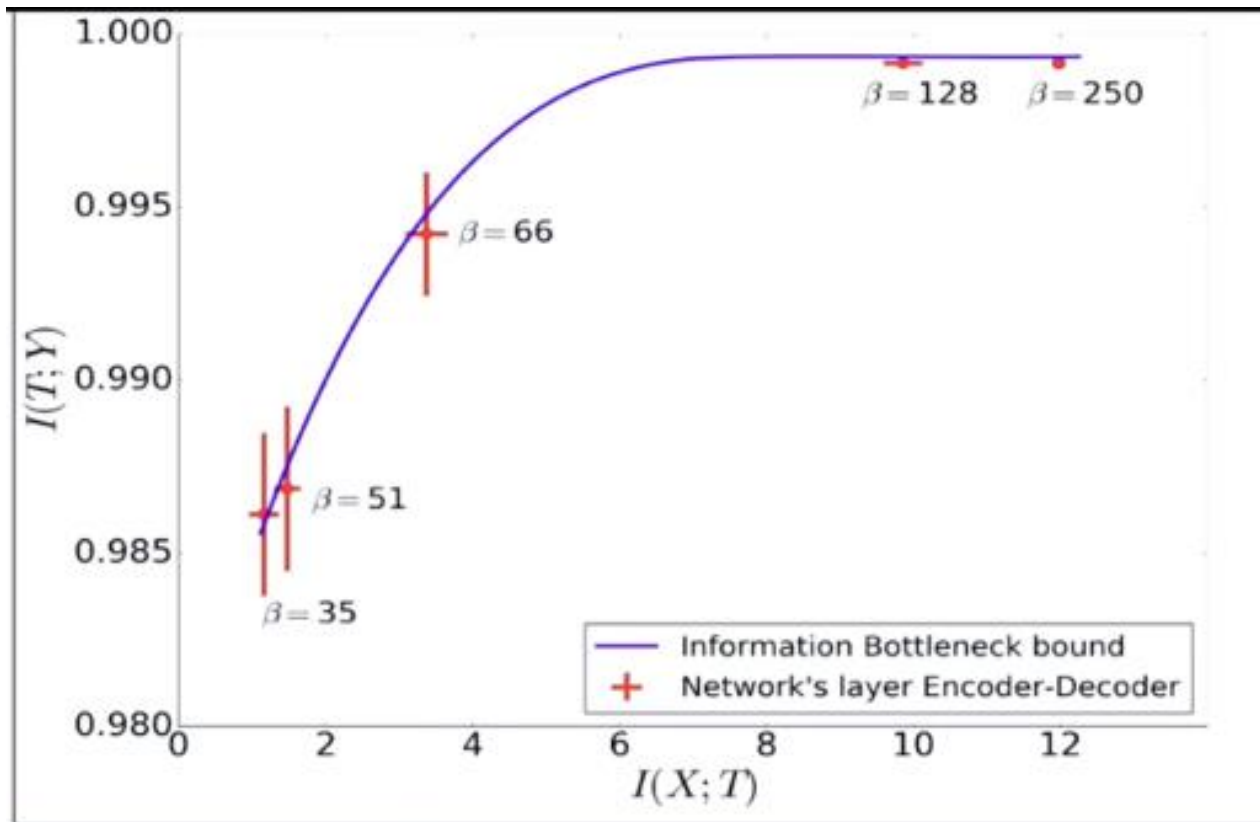
This is precisely the IB optimal encoder with $\tilde{\beta}_k$ – the encoder k -layer noise

Since $I(X; T_k) = H(X) - H(X | T_k)$, Max Entropy of the weights $\Rightarrow$ Min $I(X; T_k)$

thus SGD converges, layer by layer, to a maximally compressed representation,

which is a SUCCESSIVELY REFINEABLE APPROXIMATION of the optimal IB bound!

Optimal convergence for each layers



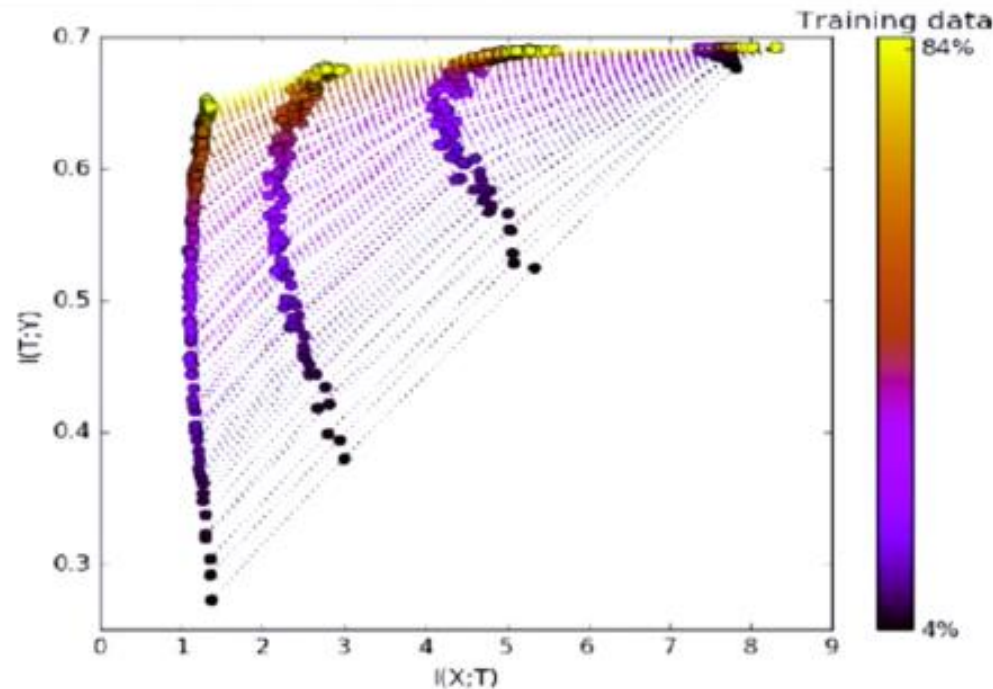
$$\beta \sim 1/\text{Variance}(\text{Noise})$$

$$\text{Variance}(\text{noise}) \sim 1/\text{size}(\text{layer})$$

Encoder Noise $\neq$ Decoder Noise

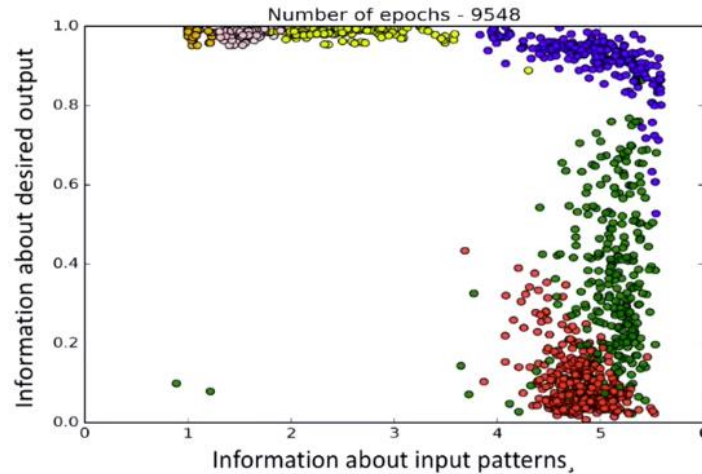
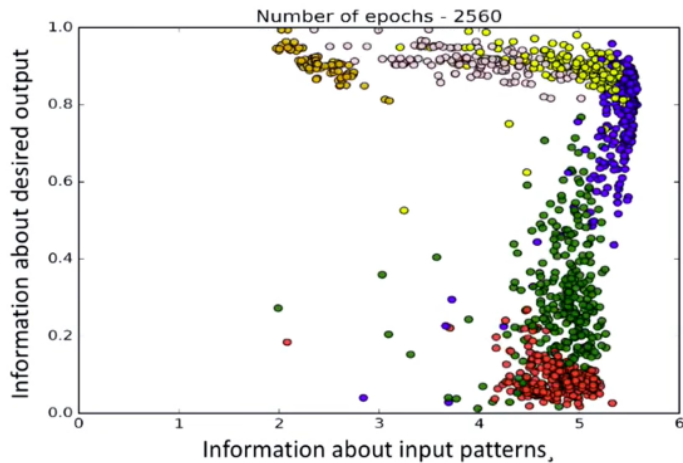
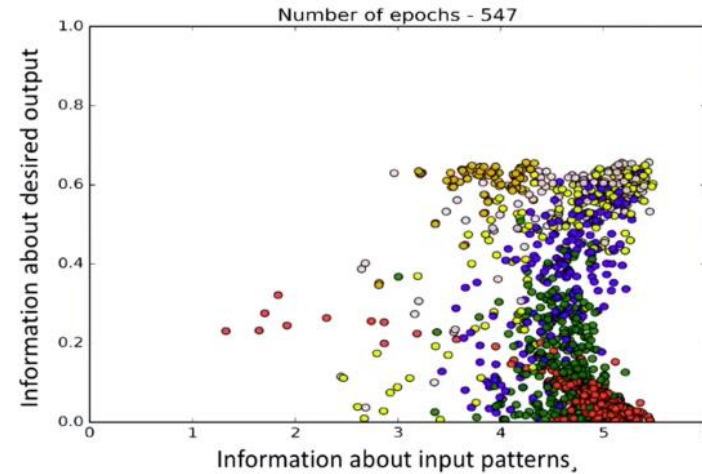
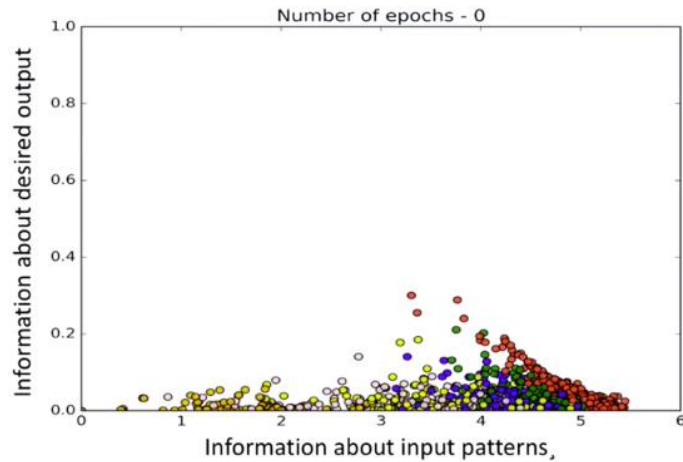
- Layers of optimal DNN converge to [a successively refineable approximation of] the optimal finite-sample IB limit information-curve
- Layers must be in “different topological phases” of the IB solutions
- The DNN encoder & decoder for each layer satisfy the IB self-consistent equations

Fixing Relative Dimension of the Layers



- Fitting larger training data require more information in the hidden layers.
- It is the **mutual-information of the last hidden layer**, which determines generalization (unlike standard hypothesis class bounds)

What Do Single Neurons represent?





The IB bifurcation points (phase-transitions)

Defining the matrices:

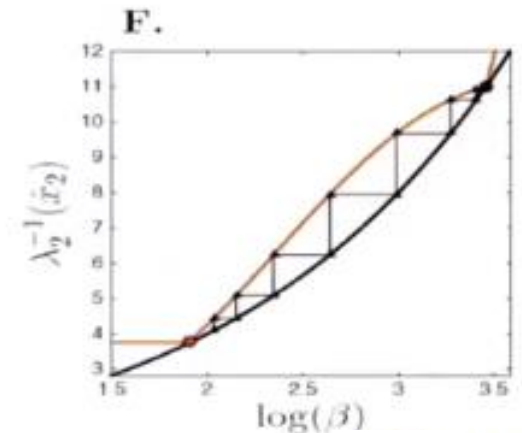
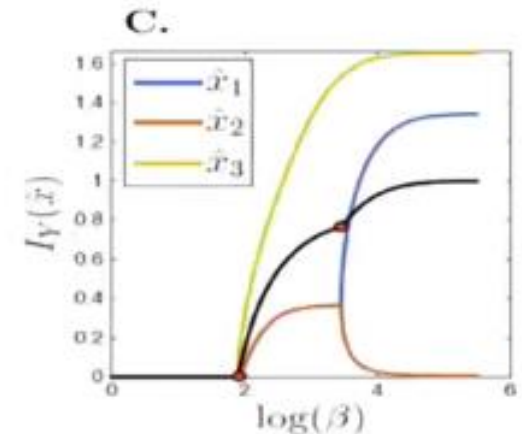
$$C_{xx'}(\hat{x}, \beta) = \sum_y \frac{p(y|x)}{p_\beta(y|\hat{x})} p_\beta(x'| \hat{x}) p(y|x')$$

$$C_{yy'}(\hat{x}, \beta) = \sum_x \frac{p(y|x)}{p_\beta(y|\hat{x})} p_\beta(x|\hat{x}) p(y'|x)$$

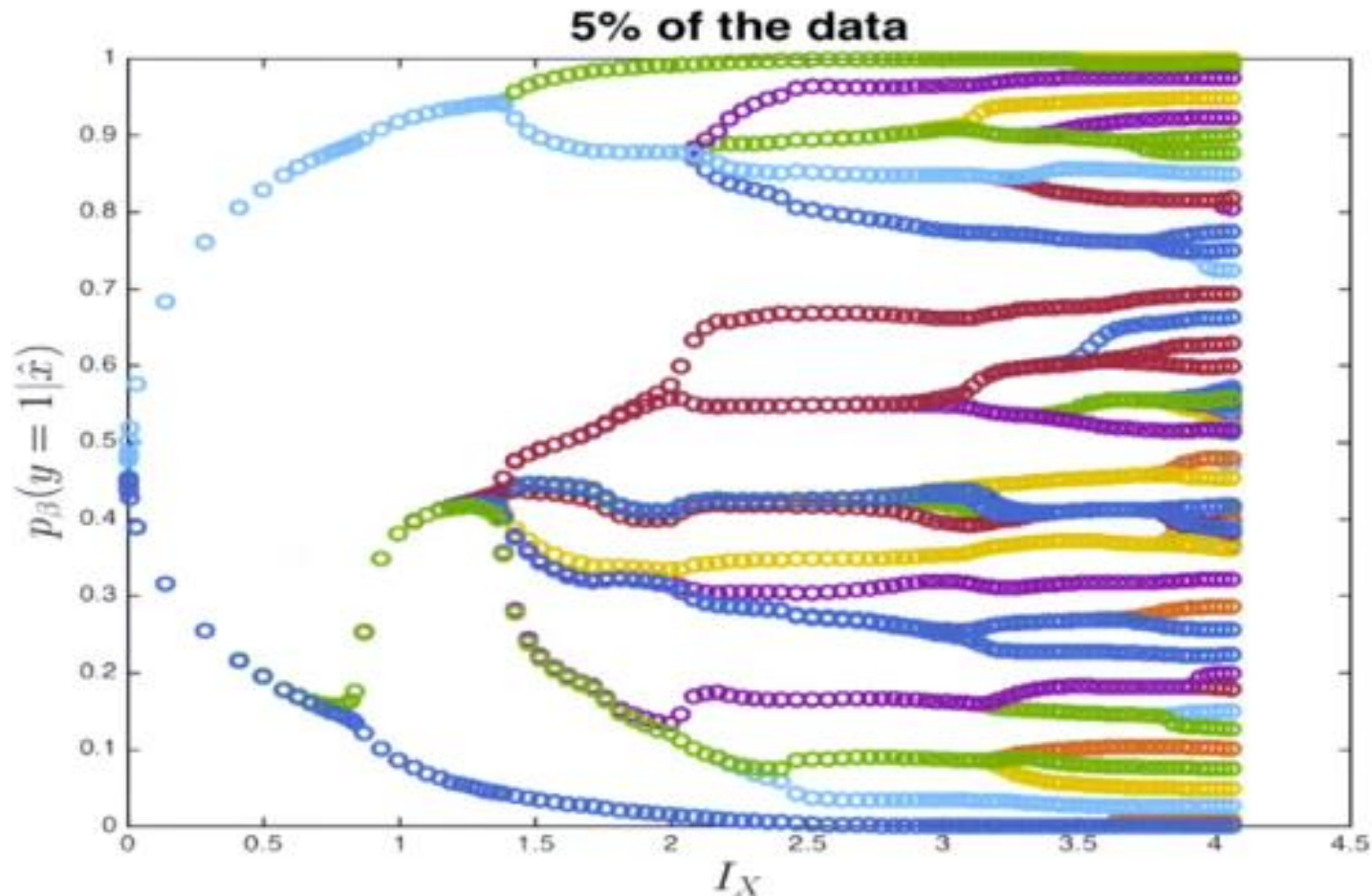
these equations can be combined into two (non-linear) eigenvalue problems:

$$\begin{aligned} [I - \beta C_{xx'}(\hat{x}, \beta)] \frac{\partial \ln p_\beta(x'|\hat{x})}{\partial \hat{x}} &= 0 \\ [I - \beta C_{yy'}(\hat{x}, \beta)] \frac{\partial \ln p_\beta(y'|\hat{x})}{\partial \hat{x}} &= 0 \end{aligned}$$

These eigenvalue problems have non-trivial solutions (eigenvectors) only at the critical bifurcation points (second order phase transitions).

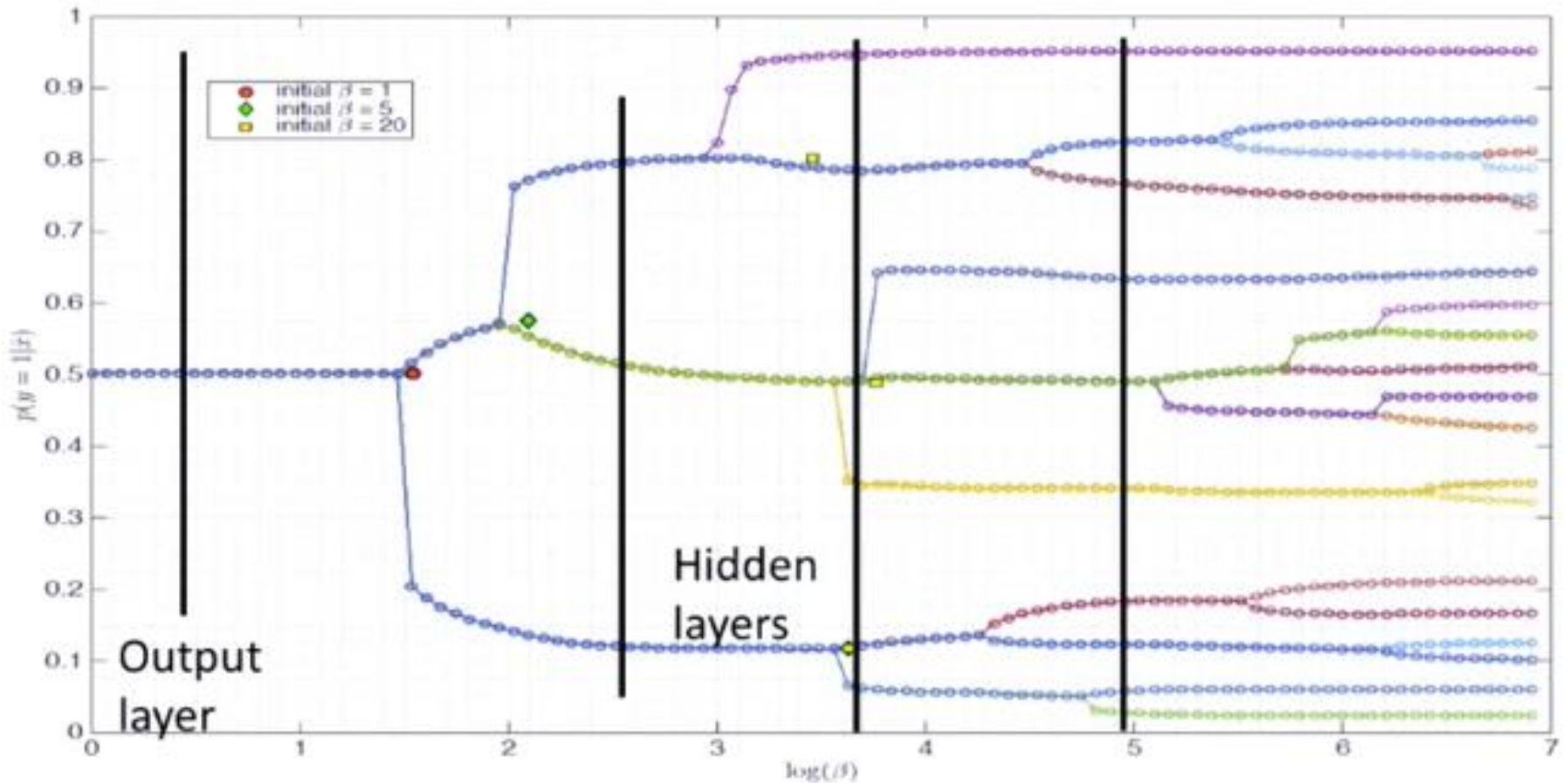


Bifurcation diagrams



In symmetric rule: layers diffusion slows down at phase transitions

Optimal Designs Principles



Topological Constraints



- **Entropy barrier can stop diffusion**
- **Phase transition occurs when split happen**
- **Topogical adjustment of the architecture**

Partial conclusion on Information Plane



- **The Information Plane provides a unique visualization of DL**
 - Most of the learning time goes to compression
 - Layers are learnt bottom up – and "help" each other
 - The layers converge to special (critical?) points on the IB bound
- **The advantage of the layers is mostly computational**
 - Relaxation times are super-linear (exponential?) in the Entropy gap
 - Hidden layers provide intermediate steps and boost convergence time
 - Hidden layers help in avoiding critical slowing down
- **Further directions**
 - Exactly solvable DNN models (through symmetry & group theory)
 - New/better learning algorithms & design principles
 - Predictions on the organization of biological layered networks .

Wavelets in Deep Learning



Why do we need to Understand NN



Szegedy, Zaremba, Sutskever, Bruna, Erhan, Goodfellow, Fergus

$$x + \epsilon = \tilde{x} \quad \text{with} \quad \|\epsilon\| < 10^{-2} \|x\|$$



correctly
classified

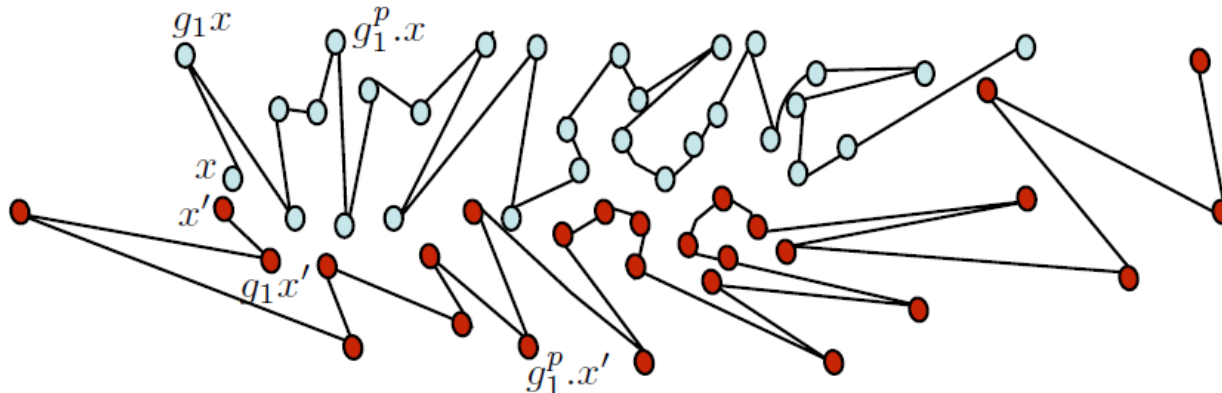
classified as
ostrich

Trial and error can guarantee reliability

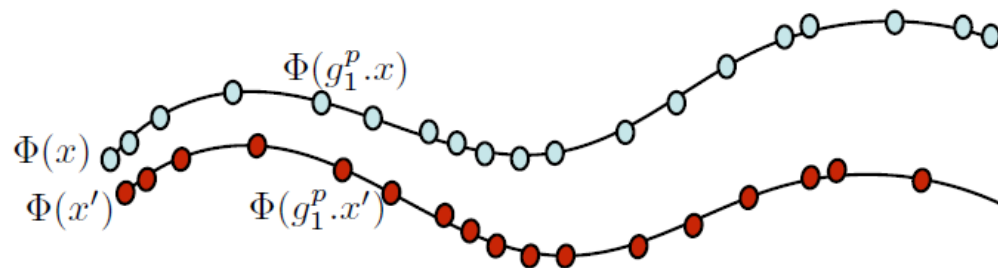
Linearizing Symmetries



- A change of variable $\Phi(x)$ must linearize the orbits $\{g.x\}_{g \in G}$



- Linearise symmetries with a change of variable $\Phi(x)$



- Lipschitz: $\forall x, g : \|\Phi(x) - \Phi(g.x)\| \leq C \|g\|$

Translations and Deformations



- Invariance to translations:

$$g.x(u) = x(u - c) \Rightarrow \Phi(g.x) = \Phi(x) .$$

- Small diffeomorphisms: $g.x(u) = x(u - \tau(u))$

Metric: $\|g\| = \|\nabla\tau\|_\infty$ maximum scaling

Linearisation by Lipschitz continuity

$$\|\Phi(x) - \Phi(g.x)\| \leq C \|\nabla\tau\|_\infty .$$

- Discriminative change of variable:

$$\|\Phi(x) - \Phi(x')\| \geq C^{-1} |f(x) - f(x')|$$

Fourier Deformation Instability



- Fourier transform $\hat{x}(\omega) = \int x(t) e^{-i\omega t} dt$

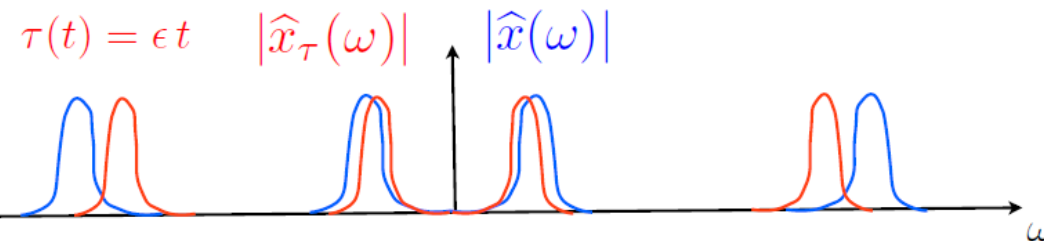
$$x_c(t) = x(t - c) \Rightarrow \hat{x}_c(\omega) = e^{-ic\omega} \hat{x}(\omega)$$

The modulus is invariant to translations:

$$\Phi(x) = |\hat{x}| = |\hat{x}_c|$$

- Instabilities to small deformations $x_\tau(t) = x(t - \tau(t))$:

$||\hat{x}_\tau(\omega)| - |\hat{x}(\omega)||$ is big at high frequencies

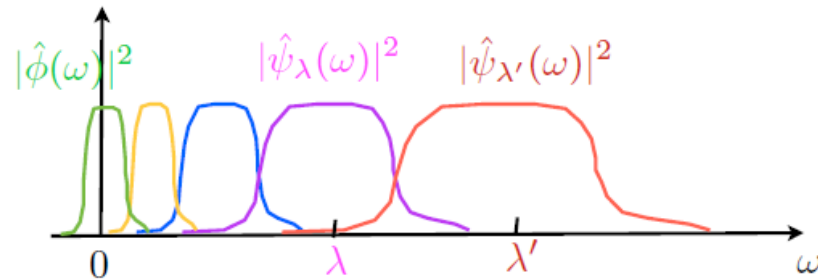
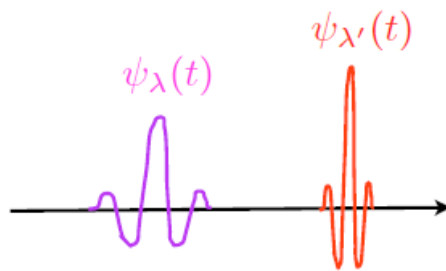


$$\Rightarrow ||\hat{x}| - |\hat{x}_\tau|| \gg \|\nabla \tau\|_\infty \|x\|$$

Multiscale Wavelets Transform



- Dilated wavelets: $\psi_\lambda(t) = 2^{-j/Q} \psi(2^{-j/Q}t)$ with $\lambda = 2^{-j/Q}$



Q-constant band-pass filters $\hat{\psi}_\lambda$

$$x \star \psi_\lambda(t) = \int x(u) \psi_\lambda(t - u) du \Rightarrow \widehat{x \star \psi_\lambda}(\omega) = \hat{x}(\omega) \hat{\psi}_\lambda(\omega)$$

- Wavelet transform: $Wx = \begin{pmatrix} x \star \phi_{2^J}(t) \\ x \star \psi_\lambda(t) \end{pmatrix}_{\lambda \leq 2^J}$: average
: higher frequencies

Preserves norm: $\|Wx\|^2 = \|x\|^2$.

Why Wavelets ?



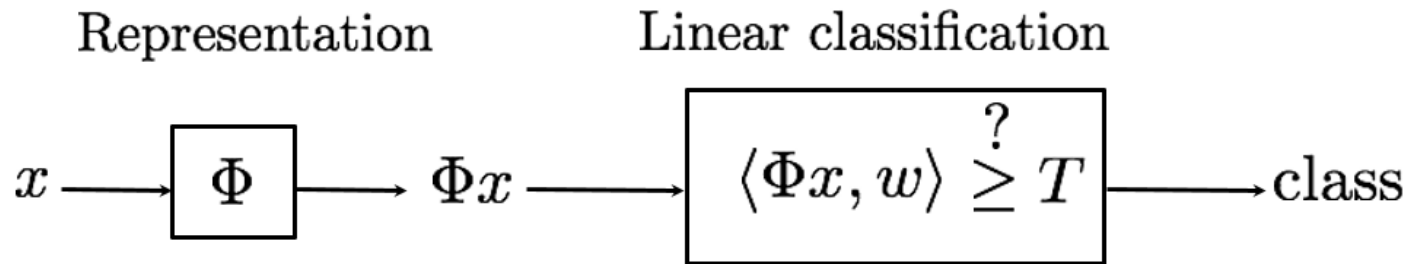
- Wavelets are uniformly stable to deformations:

if $\psi_{\lambda,\tau}(t) = \psi_{\lambda}(t - \tau(t))$ then

$$\|\psi_{\lambda} - \psi_{\lambda,\tau}\| \leq C \sup_t |\nabla \tau(t)| .$$

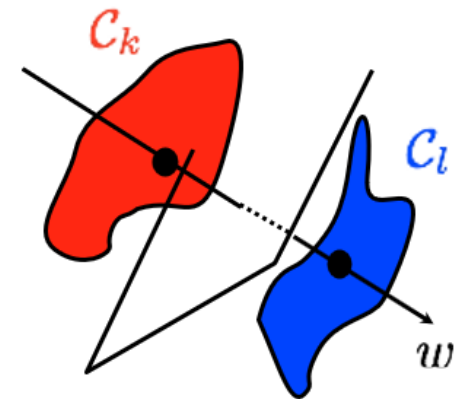
- Wavelets separate multiscale information.
- Wavelets provide sparse representations.

Classification with invariants



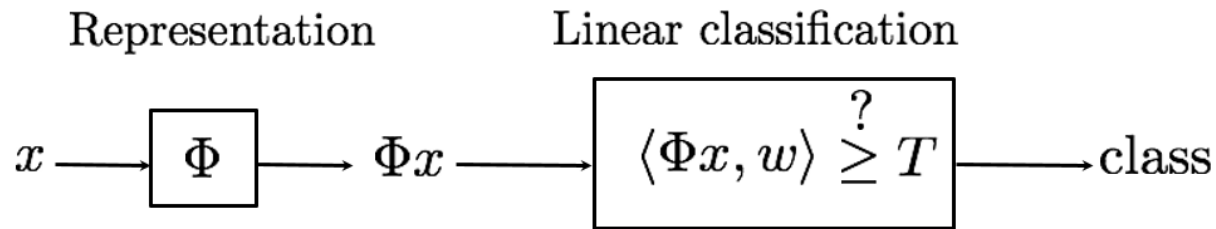
For any two classes $\mathcal{C}_k$ and $\mathcal{C}_l$ finds w so that

$$\langle \Phi x, w \rangle = \sum_n w_n \Phi_n x$$

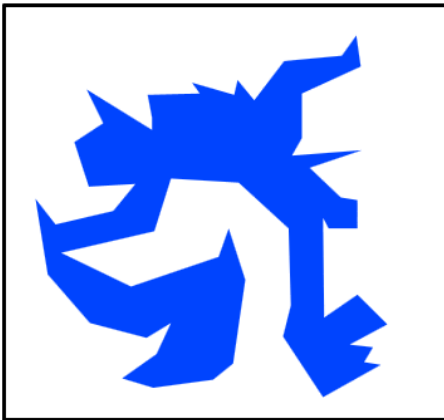


is nearly invariant and different in any $\mathcal{C}_k$ and $\mathcal{C}_l$.

Classification with Invariants



Unsupervised learning: optimizes Φ from
unlabeled **data distribution**



Alternative Models:

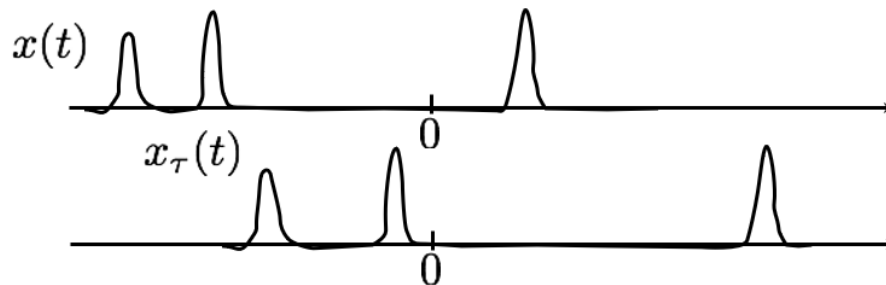
- Gaussian mixture models (clustering)
- Graphical/Markov/Manifold models

Stable Translation Invariants



- **Invariance** to translations $x_c(t) = x(t - c)$

$$\forall c \in \mathbf{R} \quad , \quad \Phi(x_c) = \Phi(x) \quad .$$



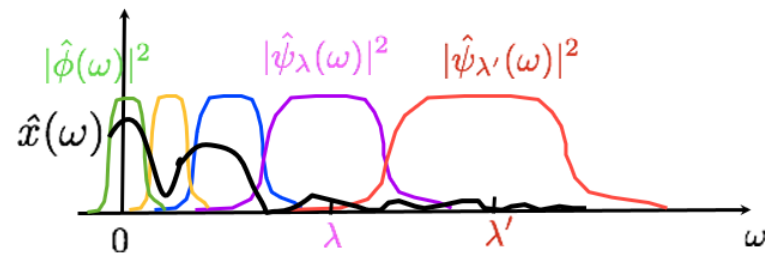
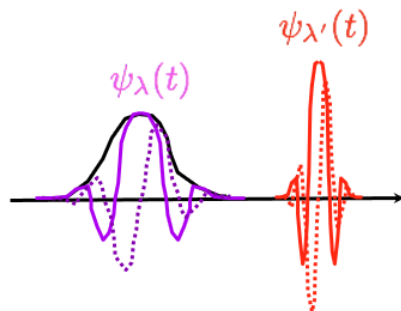
- **Lipschitz stable** to deformations $x_\tau(t) = x(t - \tau(t))$
small deformations of $x \implies$ small modifications of $\Phi(x)$

$$\forall \tau \quad , \quad \|\Phi(x_\tau) - \Phi(x)\| \leq C \underbrace{\sup_t |\nabla \tau(t)|}_{\text{deformation size}} \|x\| \quad .$$

Wavelets Transform



- Complex wavelet: $\psi(t) = \psi^a(t) + i \psi^b(t)$
- Dilated: $\psi_\lambda(t) = 2^{-j} \psi(2^{-j}t)$ with $\lambda = 2^{-j}$.



- Wavelet transform: $x \star \psi_\lambda(t) = \int x(u) \psi_\lambda(t - u) du$

$$Wx = \begin{pmatrix} x \star \phi(t) \\ x \star \psi_\lambda(t) \end{pmatrix}_{t,\lambda}$$

Unitary: $\|Wx\|^2 = \|x\|^2$.

Why Wavelets ?

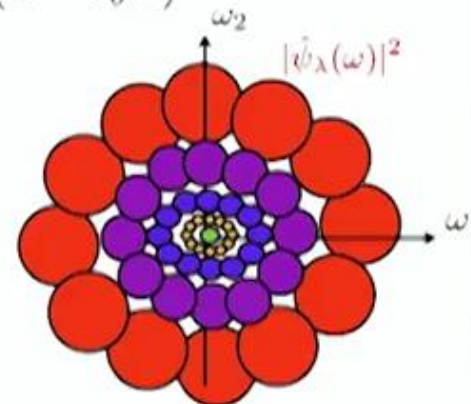
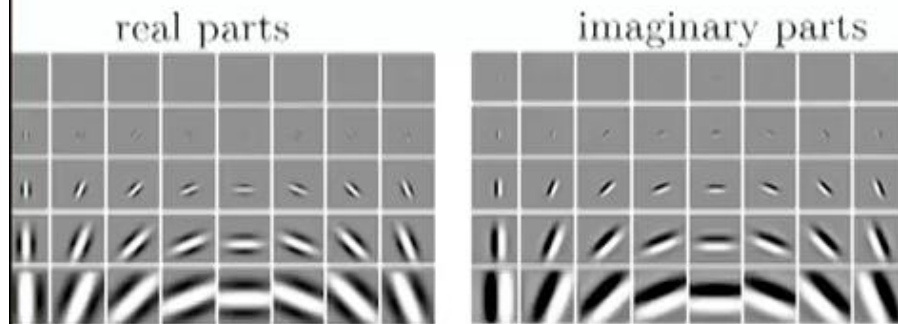


- The wavelet dictionary $\{\psi_\lambda(t - u)\}_{t,\lambda}$ is translation invariant.
- Wavelets are uniformly stable to deformations:
if $\psi_{\lambda,\tau}(t) = \psi_\lambda(t - \tau(t))$ then
$$\|\psi_\lambda - \psi_{\lambda,\tau}\| \leq C \sup_t |\tau'(t)| .$$
- Provides sparse representations...

Scale separation with Wavelets

- Wavelet filter $\psi(u)$: 

rotated and dilated: $\psi_{2^j, \theta}(u) = 2^{-j} \psi(2^{-j} r_\theta u)$



- Wavelet transform: $Wx = \begin{pmatrix} x \star \phi_{2^J}(u) \\ x \star \psi_{2^j, \theta}(u) \end{pmatrix}_{j \leq J, \theta}$: average
: higher frequencies

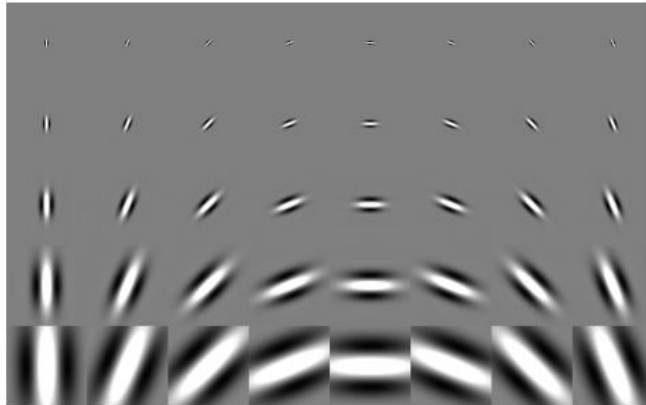
Preserves norm: $\|Wx\|^2 = \|x\|^2$.

Image Wavelet Transform

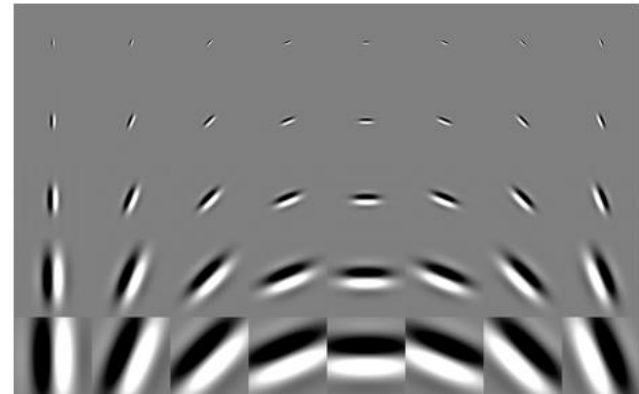


- Complex wavelet: $\psi(t) = \psi^a(t) + i \psi^b(t)$, $t = (t_1, t_2)$
rotated and dilated: $\psi_\lambda(t) = 2^{-j} \psi(2^{-j} r t)$ with $\lambda = (2^j, r)$

real parts



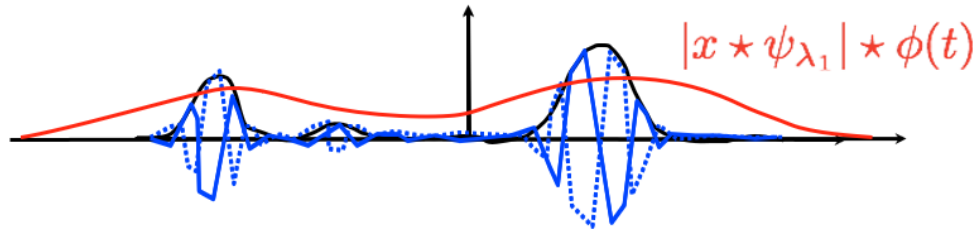
imaginary parts



- Wavelet transform: $Wx = \begin{pmatrix} x \star \phi(t) \\ x \star \psi_\lambda(t) \end{pmatrix}_{t,\lambda}$

Unitary: $\|Wx\|^2 = \|x\|^2$.

Wavelets Translation Invariance

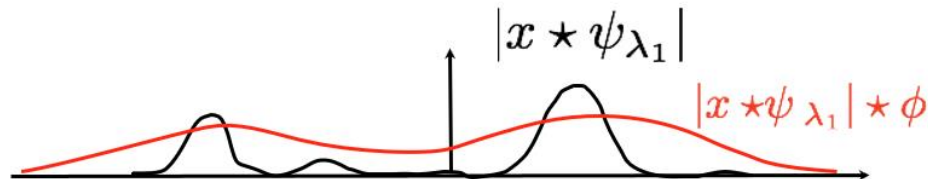


- The modulus $|x \star \psi_{\lambda_1}|$ is a regular envelop
- The average $|x \star \psi_{\lambda_1}| \star \phi(t)$ is invariant to small translations relatively to the support of ϕ .
- Full translation invariance at the limit:

$$\lim_{\phi \rightarrow 1} |x \star \psi_{\lambda_1}| \star \phi(t) = \int |x \star \psi_{\lambda_1}(u)| du = \|x \star \psi_{\lambda_1}\|_1$$

but few invariants.

Recovering Lost Information



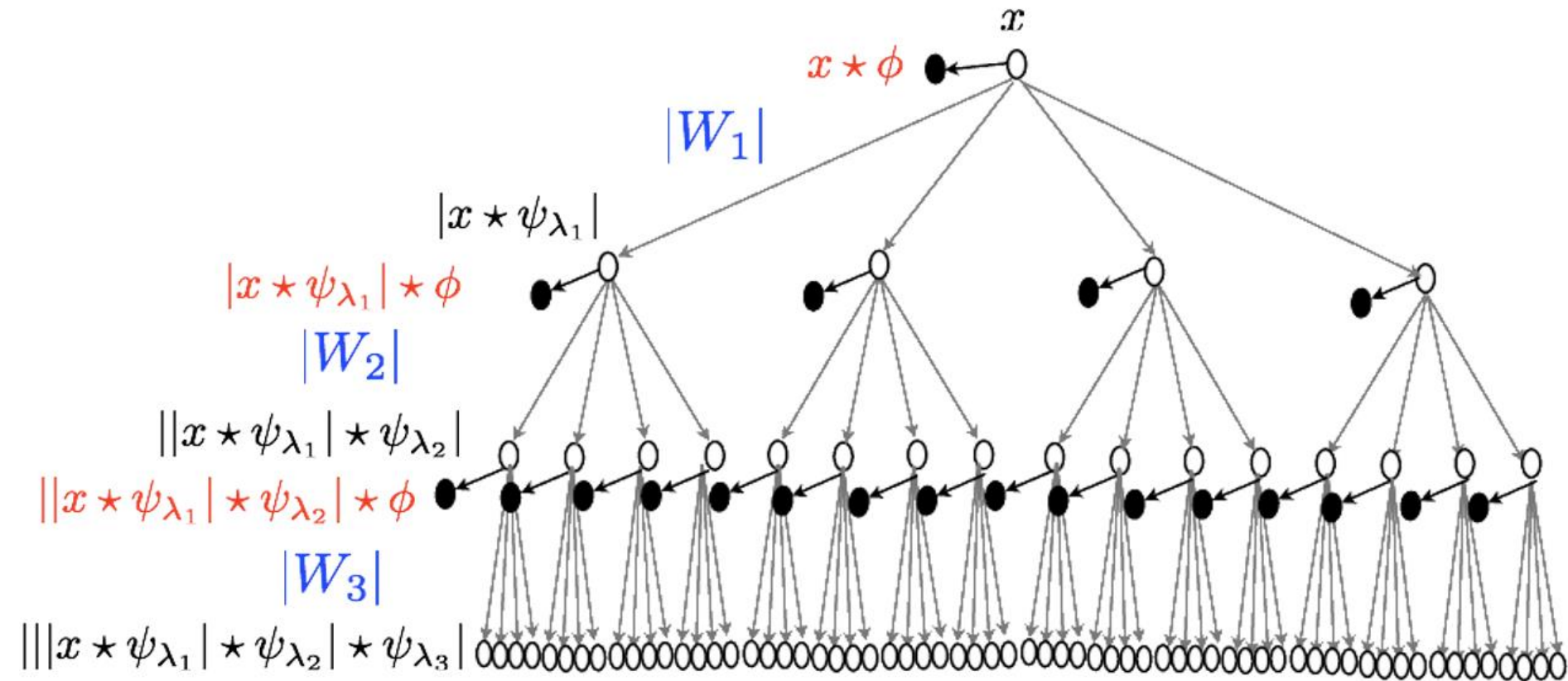
- The high frequencies of $|x \star \psi_{\lambda_1}|$ are in wavelet coefficients:

$$W|x \star \psi_{\lambda_1}| = \begin{pmatrix} |x \star \psi_{\lambda_1}| \star \phi(t) \\ |x \star \psi_{\lambda_1}| \star \psi_{\lambda_2}(t) \end{pmatrix}_{t, \lambda_2}$$

- Translation invariance by time averaging the amplitude:

$$\forall \lambda_1, \lambda_2, \quad ||x \star \psi_{\lambda_1}| \star \psi_{\lambda_2}| \star \phi(t)$$

Deep Convolution Network



Scattering Vector



Network output:

$$Sx = \begin{pmatrix} x \star \phi(u) \\ |x \star \psi_{\lambda_1}| \star \phi(u) \\ ||x \star \psi_{\lambda_1}| \star \psi_{\lambda_2}| \star \phi(u) \\ |||x \star \psi_{\lambda_2}| \star \psi_{\lambda_2}| \star \psi_{\lambda_3}| \star \phi(u) \\ \dots \end{pmatrix}_{u, \lambda_1, \lambda_2, \lambda_3, \dots}$$

Scattering Properties



$$Sx = \begin{pmatrix} x \star \phi(u) \\ |x \star \psi_{\lambda_1}| \star \phi(u) \\ ||x \star \psi_{\lambda_1}| \star \psi_{\lambda_2}| \star \phi(u) \\ |||x \star \psi_{\lambda_2}| \star \psi_{\lambda_2}| \star \psi_{\lambda_3}| \star \phi(u) \\ \dots \end{pmatrix}_{u, \lambda_1, \lambda_2, \lambda_3, \dots}$$

Theorem: *For appropriate wavelets, a scattering is*

contractive $\|Sx - Sy\| \leq \|x - y\|$

preserves norms $\|Sx\| = \|x\|$

stable to deformations $x_\tau(t) = x(t - \tau(t))$

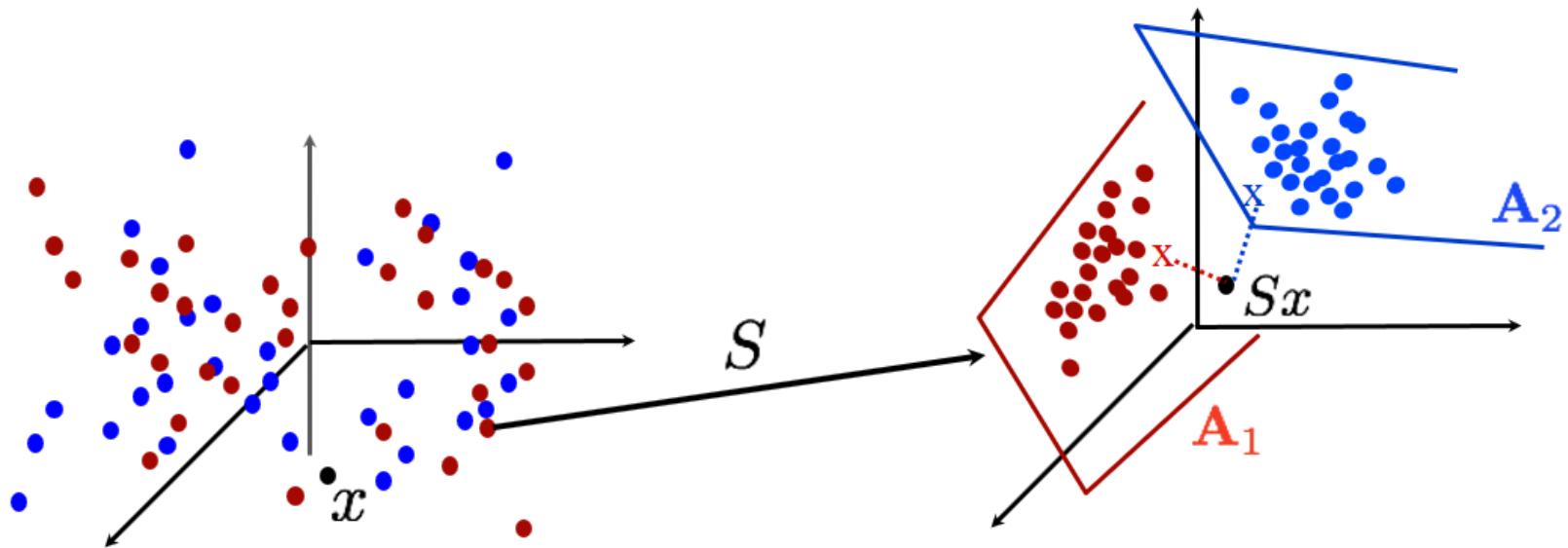
$$\|Sx - Sx_\tau\| \leq C \sup_t |\nabla \tau(t)| \|x\|$$

$\Rightarrow$ linear discriminative classification from $\Phi x = Sx$

Affine Space Classification



- Each class $\mathcal{C}_k$ is approximated by an affine space $\mathbf{A}_k$ in the scattering domain, computed with a PCA.



Digit Classification : MNIST

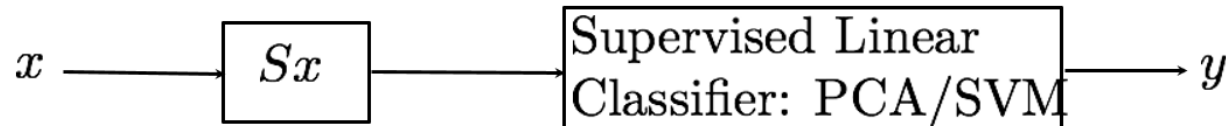


3 6 8 1 7 9 6 6 9 1
6 7 5 7 8 6 3 4 8 5

J. Bruna

Translation invariant
Stable to deformations

Learns deformation invariants



Classification Errors

Training	Conv. Net.	Scat. m = 1	Scat. m=2
60000	0.5%	0.7%	0.4 %

LeCun et. al.

Expected Scattering Transform



$$X(t) \text{ stationary process: } \bar{S}X = \begin{pmatrix} E(X) \\ E(|X \star \psi_{\lambda_1}|) \\ E(|X \star \psi_{\lambda_1}| \star \psi_{\lambda_2}) \\ E(|X \star \psi_{\lambda_2}| \star \psi_{\lambda_3}) \\ \dots \end{pmatrix}_{\lambda_1, \lambda_2, \lambda_3, \dots}$$

Theorem: *A scattering is*

$$\text{contractive } \|\bar{S}X - \bar{S}Y\|^2 \leq E(|X - Y|^2)$$

$$\text{preserves norms } \|\bar{S}X\|^2 = E(|X|^2)$$

$$\text{stable to stationary deformations } X_\tau(t) = X(t - \tau(t))$$

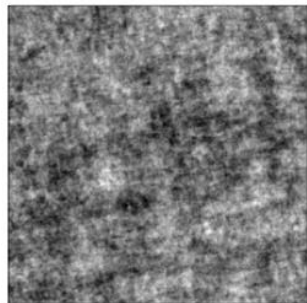
$$\|\bar{S}X - \bar{S}X_\tau\| \leq C \sup_t |\nabla \tau(t)| E(|X|^2)^{1/2} .$$

Texture with the same spectrum

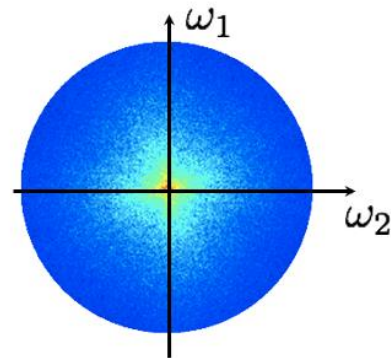
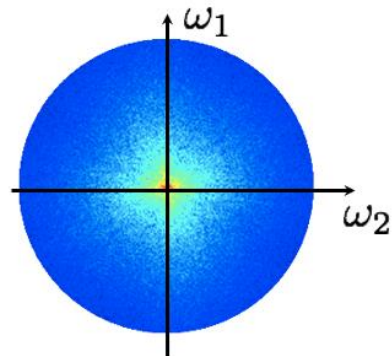


$x(t)$: stationary process

Textures
 $x(t)$

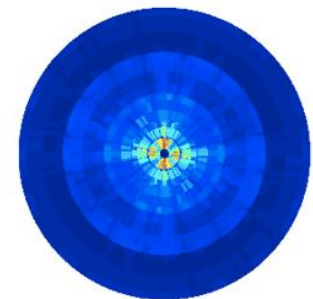
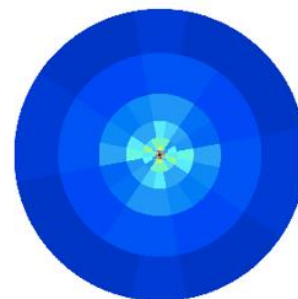
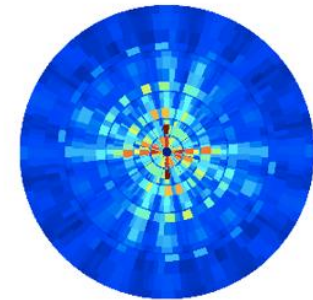
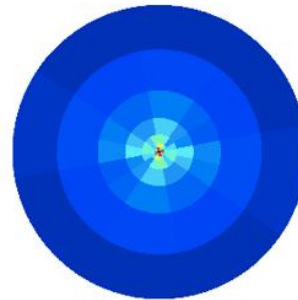


Fourier
Power Spectrum



Wavelet Scattering

$$|x \star \psi_{\lambda_1}| \star \phi \quad ||x \star \psi_{\lambda_1}| \star \psi_{\lambda_2}| \star \phi$$



window size = image size

Representation of Random Processes



- An expected scattering is a non-complete representation

$$\bar{S}X = \left(\begin{array}{rcl} E(X) & = & E(U_0X) \\ E(|X \star \psi_{\lambda_1}|) & = & E(U_1X) \\ E(||X \star \psi_{\lambda_1}| \star \psi_{\lambda_2}|) & = & E(U_2X) \\ E(|||X \star \psi_{\lambda_2}| \star \psi_{\lambda_2}| \star \psi_{\lambda_3}|) & = & E(U_3X) \\ \dots & & \end{array} \right)_{\lambda_1, \lambda_2, \lambda_3, \dots}$$

Theorem (Boltzmann) The distribution $p(x)$ which satisfies

$$\int_{\mathbb{R}^N} U_m x p(x) dx = E(U_m X)$$

and maximizes the entropy $-\int p(x) \log p(x) dx$

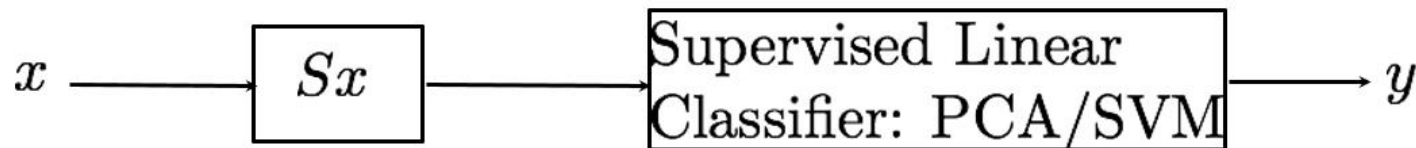
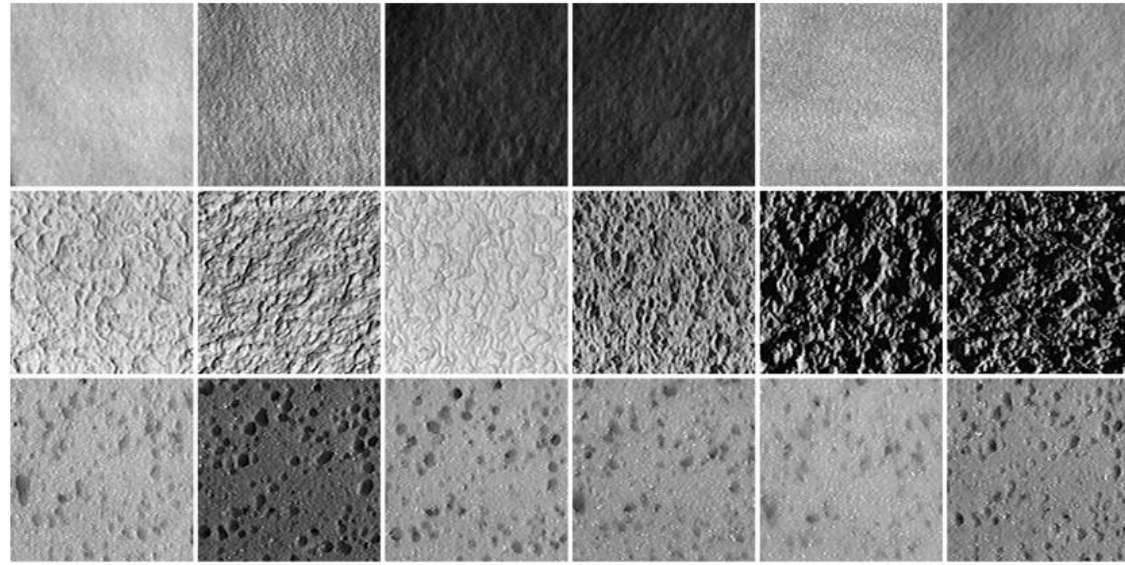
can be written: $p(x) = \frac{1}{Z} \exp \left(\sum_{m=1}^{\infty} \lambda_m \cdot U_m x \right)$

Classification of Textures



J. Bruna

CUREt database
61 classes



Training per class	Fourier Spectr.	Histogr. Features	Scattering
46	1%	1%	0.2 %

Wavelet Transform on a Group



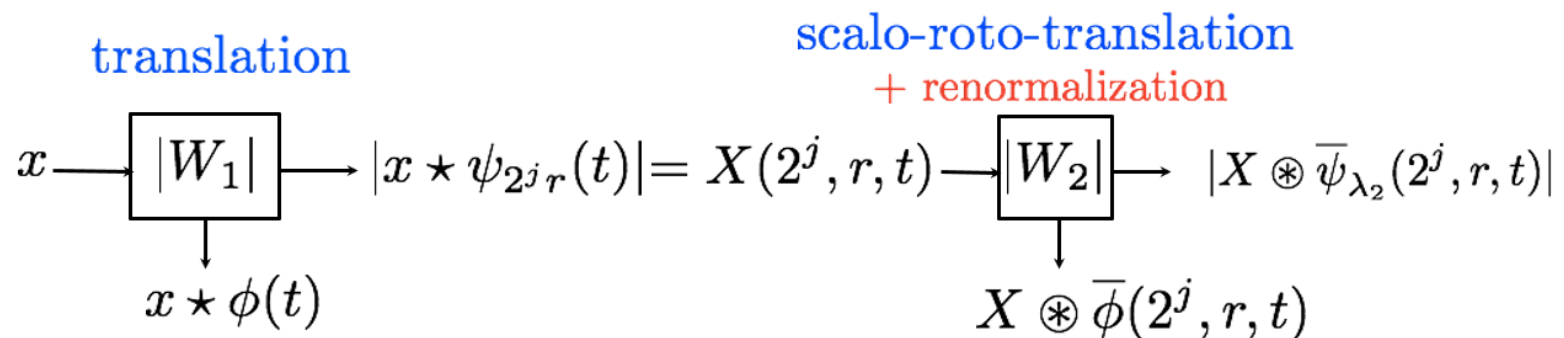
Laurent Sifre

- Roto-translation group $G = \{g = (r, t) \in SO(2) \times \mathbb{R}^2\}$

$$(r, t) \cdot x(u) = x(r^{-1}(u - t))$$

- Averaging on G : $X \circledast \bar{\phi}(g) = \int_G X(g') \bar{\phi}(g'^{-1}g) dg'$

- Wavelet transform on G : $W_2 X = \begin{pmatrix} X \circledast \bar{\phi}(g) \\ X \circledast \bar{\psi}_{\lambda_2}(g) \end{pmatrix}_{\lambda_2, g}$.

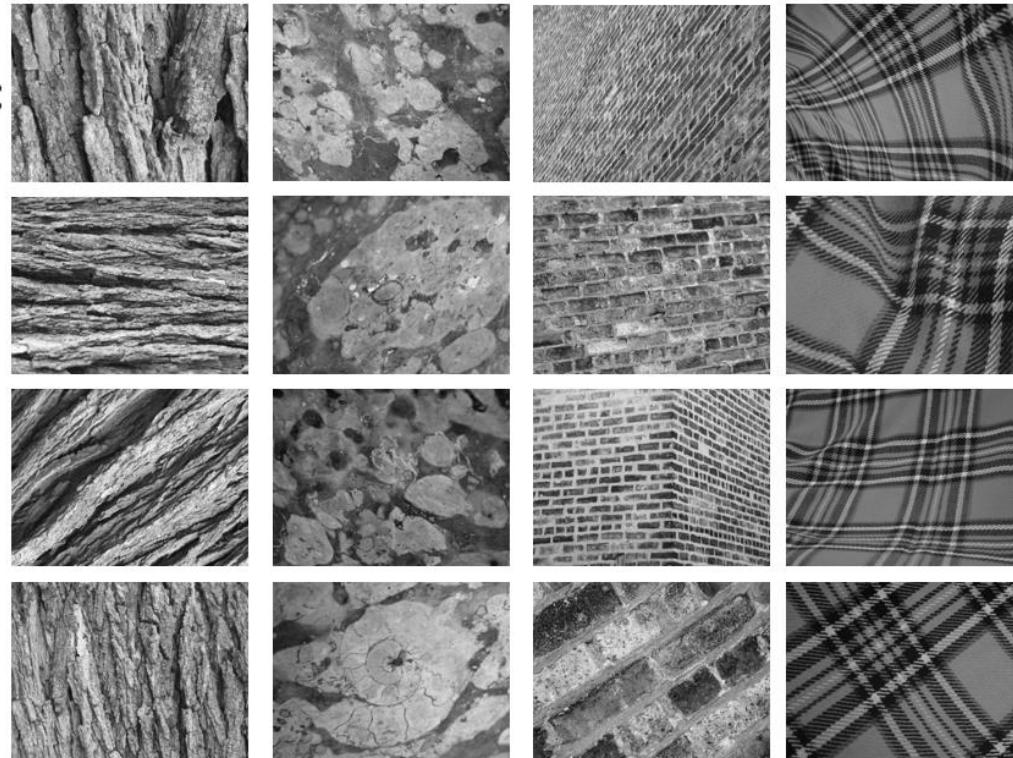


Rotation and Scaling Invariance



Laurent Siffr

UIUC database:
25 classes



Scattering classification errors

Training	Translation	Transl + Rotation	+ Scaling
20	20 %	2%	0.6%

Conclusion on the Use of Wavelets



- Deep neural networks can be analyzed mathematically
- High dimensional stochastic models through contractions
- Unsupervised learning optimized with sparse contractions
- Papers and softwares: www.di.ens.fr/data/software/scatnet/

Renormalization



Renormalization



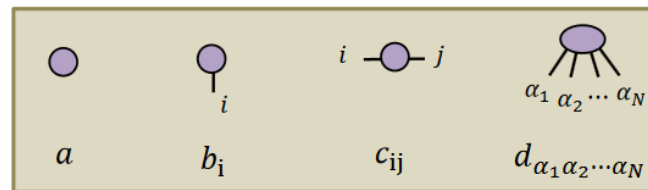
$$|\Psi\rangle = \sum_{i_1, i_2, \dots, i_N} \Psi_{i_1 i_2 \dots i_N} |i_1 i_2 \dots i_N\rangle$$

2^N
parameters

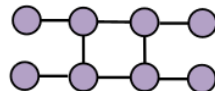
↓

tensor network

graphical
notation

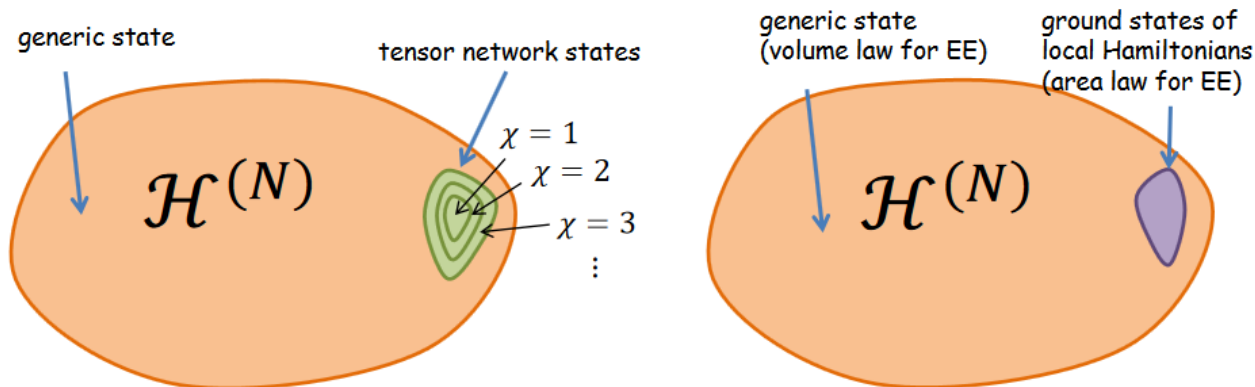
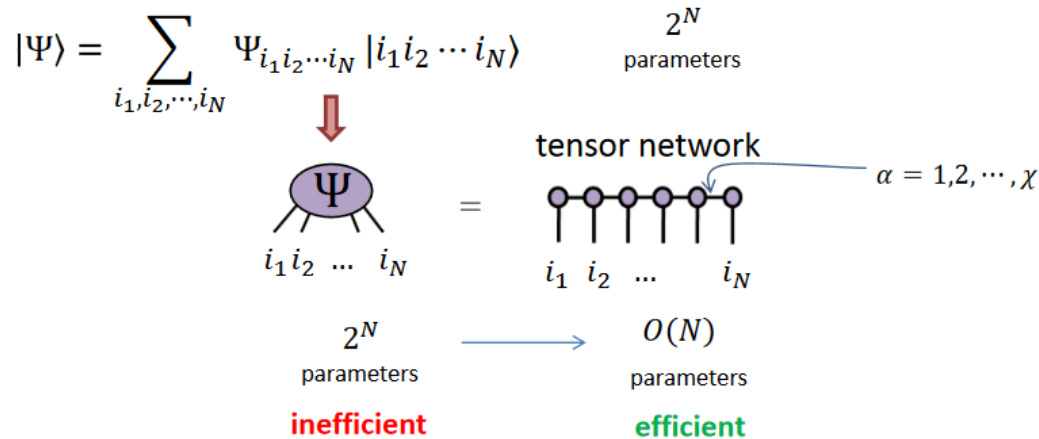


why bother?



$$\sum \dots A_{ijk} B_{jlm} C_{nko} D_{kmr} x_i y_l z_n v_r$$

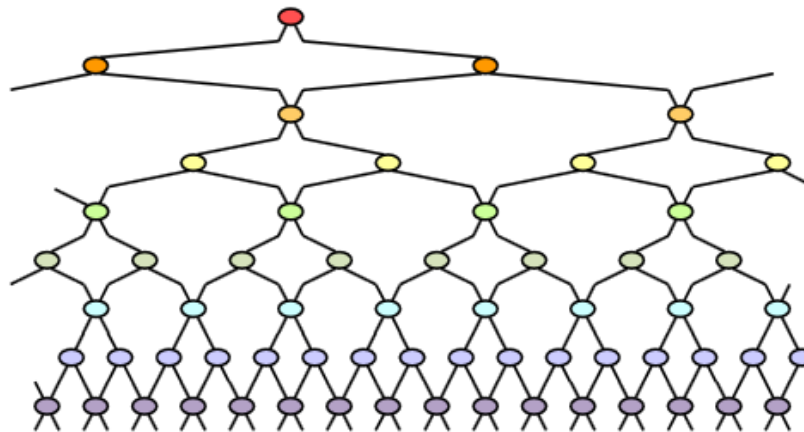
Many Body Wave-Function of N Spins



Example of Tensor Network



Multi-scale entanglement renormalization ansatz (MERA)

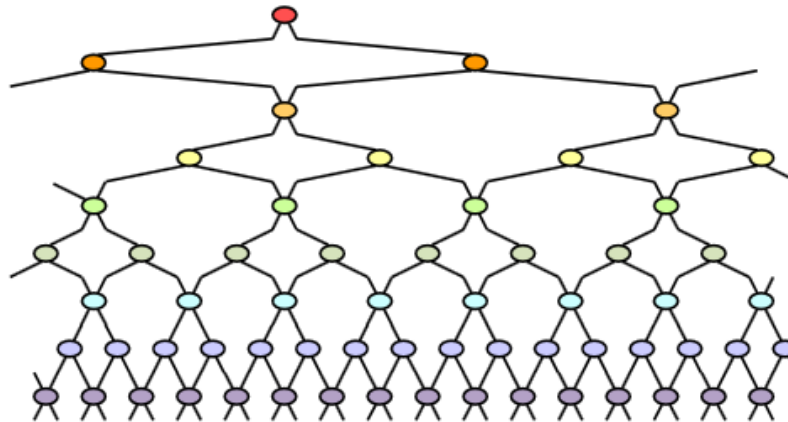


- Variational class of states for 1d systems, which extends in space and scale
- Variational parameters for different length scales stored in different tensors
- It is secretly a **quantum circuit** and an **RG transformation**

Exemple of Tensor Network



Multi-scale entanglement renormalization ansatz
(MERA)

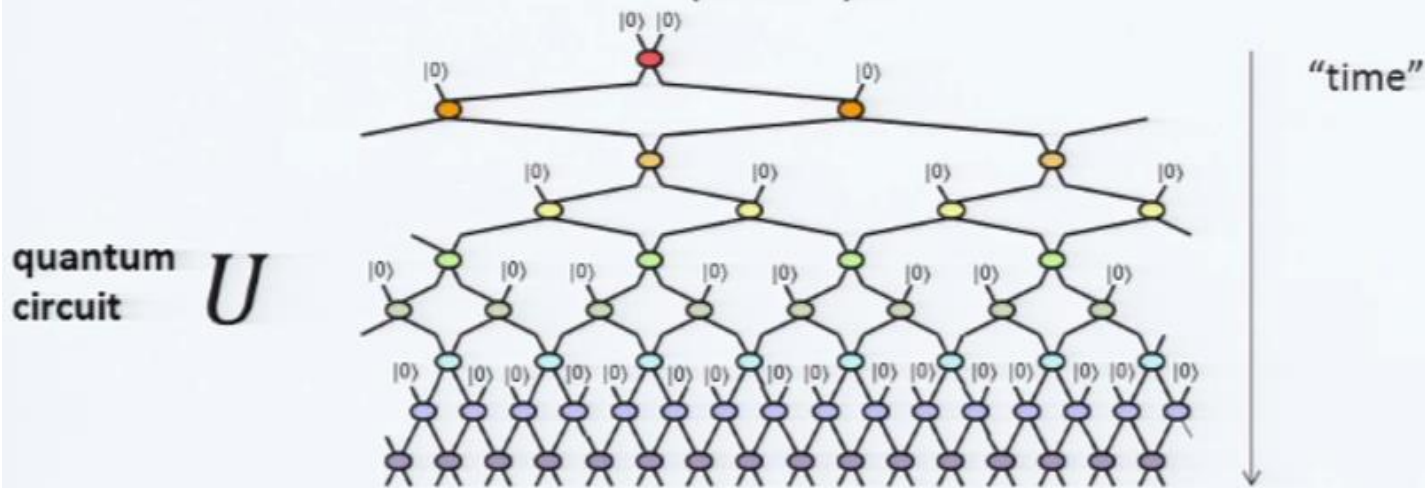


- Variational class of states for 1d systems, which extends in space and scale
- Variational parameters for different length scales stored in different tensors
- It is secretly a **quantum circuit** and an **RG transformation**

Multi Scale Entanglement Renormalization



Multi-scale entanglement renormalization ansatz (MERA)



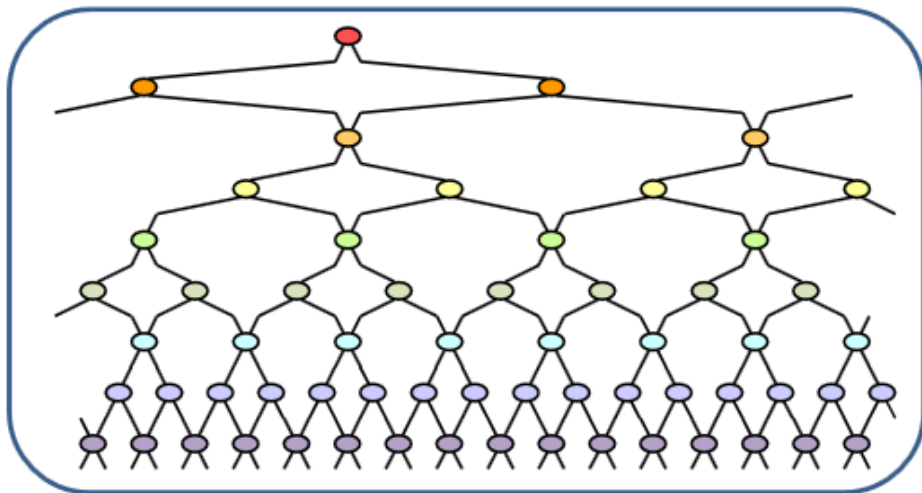
ground state $|\Psi\rangle = U |0\rangle \otimes |0\rangle \otimes \dots \otimes |0\rangle$

Entanglement is introduced by the gates at different times (=scales)

Multi Scale Entanglement Renormalization

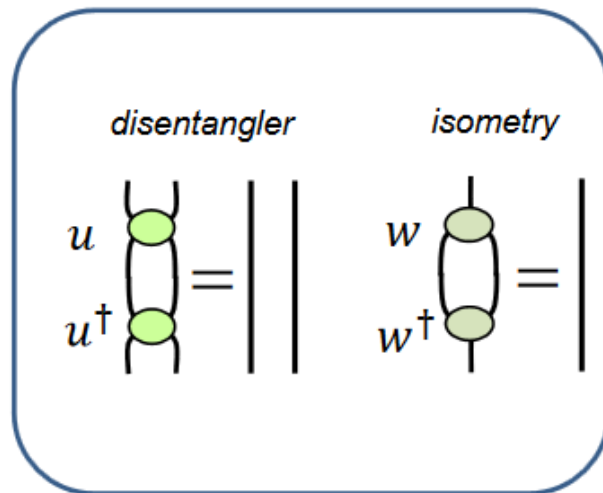


MERA = tensor network + isometric/unitary constraints



~ hyperbolic plane?

(Swingle 2009)



~ de Sitter space?

(Beny 2011, Czech 2015)



Causal structure

essential for many MERA properties
and computational efficiency

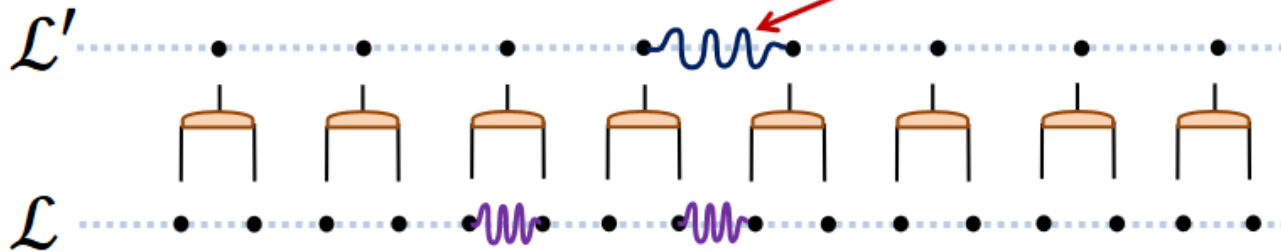
Entanglement Renormalization



Kadanoff (1966)
blocking

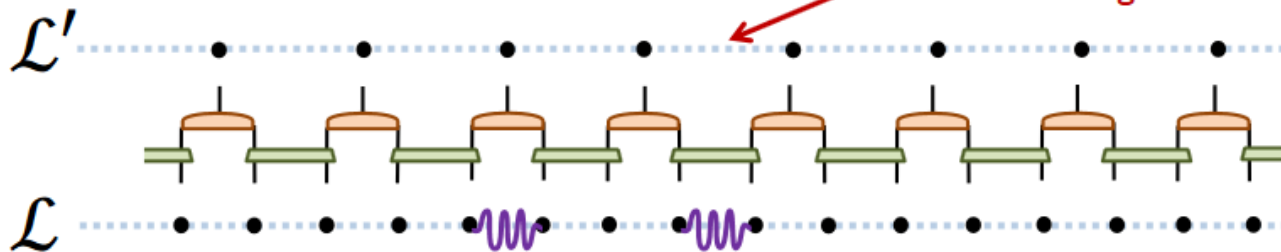
+ White (1992)
variational optimization

failure to remove
some short-range
entanglement !

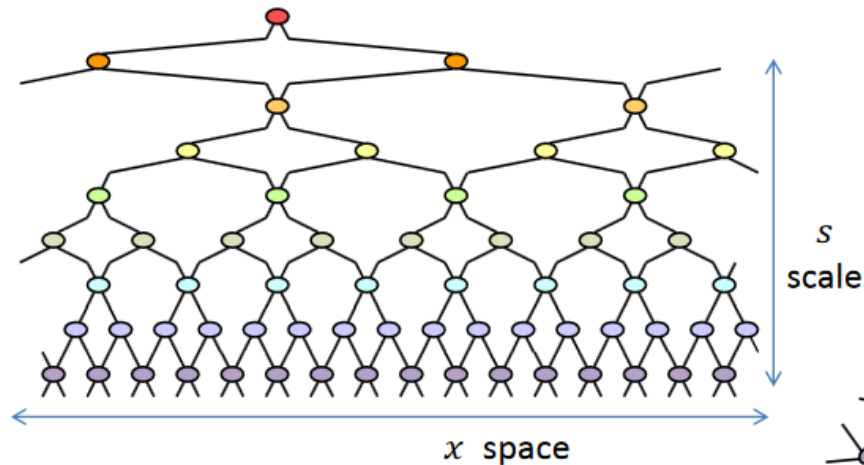


Entanglement renormalization (2005)

removal of *all*
short-range
entanglement



RG Flows in the space of hamiltonians



- entanglement entropy

$$S_L \approx \log(L)$$

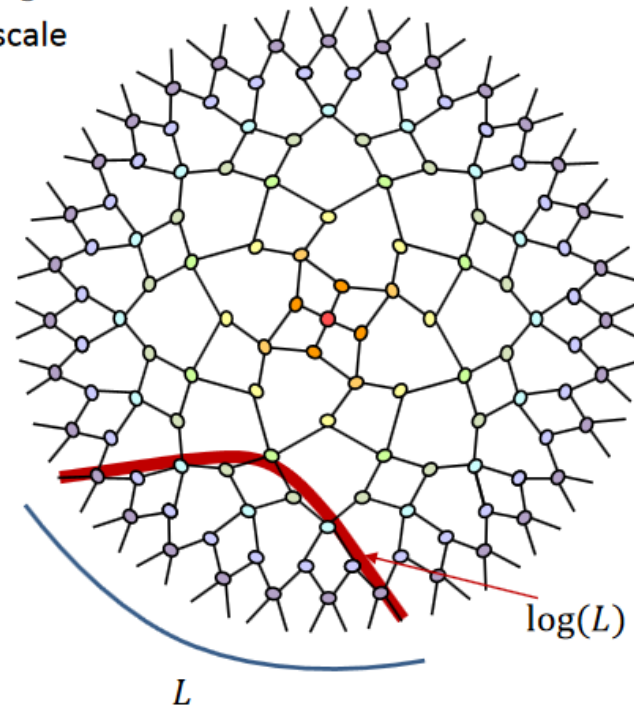
Computation of density matrix requires tracing out $\sim \log(L)$ indices

- two-point correlations

$$C(L) \approx L^{-2\Delta}$$

Geodesic distance $D \approx \log(L)$

$$C(L) \approx e^{-D} = e^{-2\Delta \log(L)} = L^{-2\Delta}$$

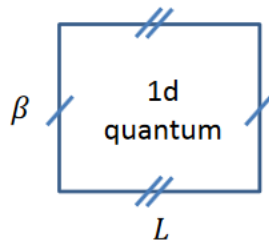


RG Flows in the space of hamiltonians



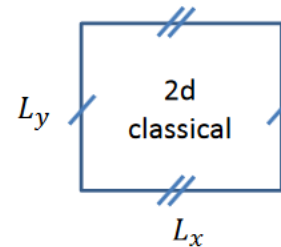
Euclidean path integral

$$Z(\lambda) = \text{tr} e^{-\beta H_q^{1d}}$$



Statistical partition function

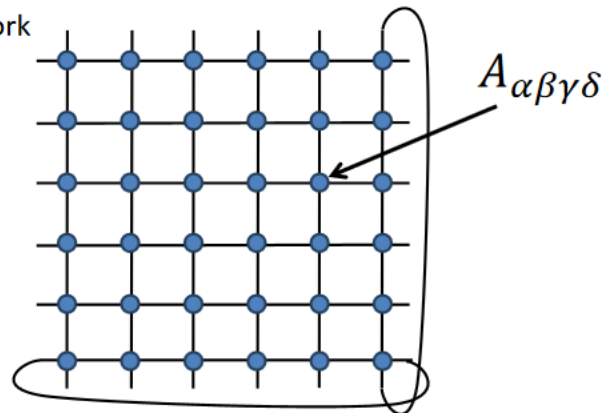
$$Z(T) = \sum_{\{s\}} e^{-\frac{1}{T} H_{cl}^{2d}}$$



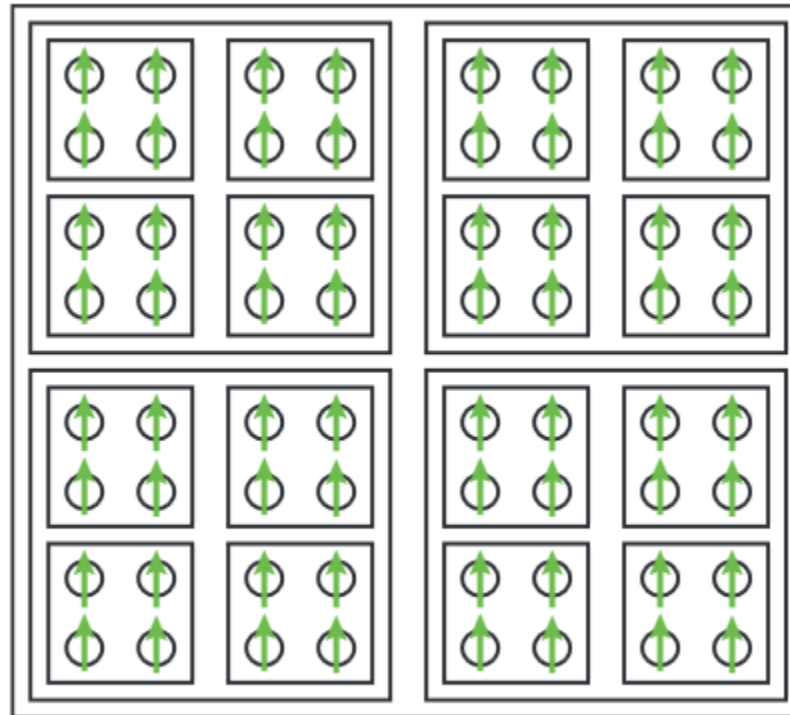
~

as a tensor network

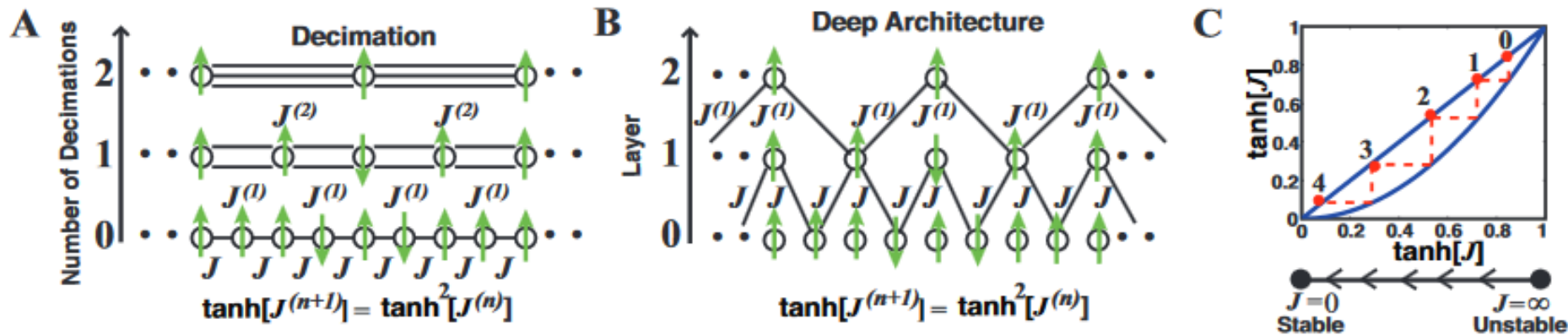
$$Z =$$



Block spin renormalization



RG and deep learning in the one-dimensional Ising Model



(A) A decimation based renormalization transformation for the ferromagnetic 1-D Ising model. At each step, half the spins are decimated, doubling the effective lattice spacing. After n successive decimations, the spins can be described using a new 1-D Ising model with a coupling $J^{(n)}$ between spins. Couplings at a given layer are related to couplings at a previous layer through the square of the hyperbolic tangent function.

(B) Decimation-based renormalization transformations can also be realized using the deep architecture where the weights between the $n+1$ and n -th hidden layer are given by $J^{(n)}$.

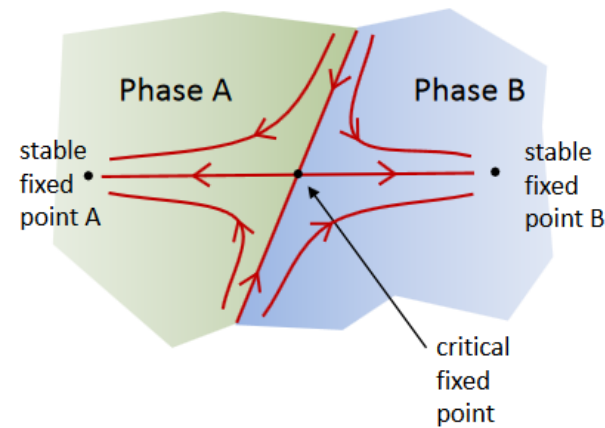
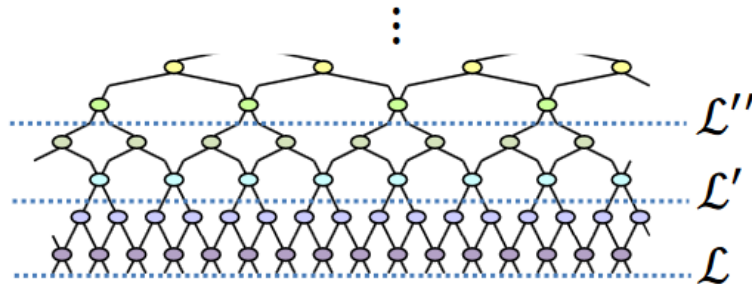
(C) Visualizing the renormalization group flow of the couplings for 1-D Ferromagnetic Ising model. Under four successive decimations or equivalently as we move up four layers in the deep architecture, the couplings (marked by red dots) get smaller. Eventually, the couplings are attracted to stable fixed point $J=0$.

RG Flows in the space of hamiltonians



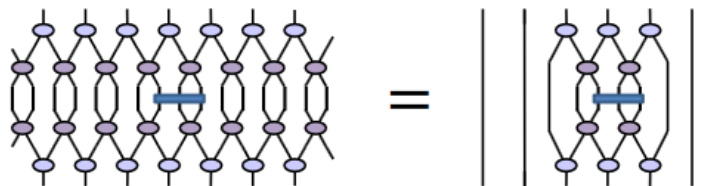
MERA defines an RG flow
in the space of wave-functions

$$|\Psi\rangle \rightarrow |\Psi'\rangle \rightarrow |\Psi''\rangle \rightarrow \dots$$



... and in the space of Hamiltonians

$$H \rightarrow H' \rightarrow H'' \rightarrow \dots$$

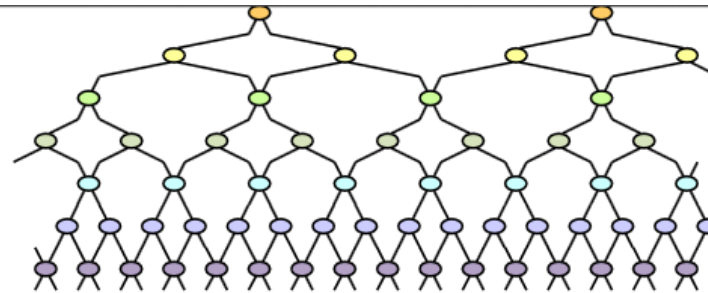


local operators
are mapped into
local operators !

Summary



MERA



- Variational parameters for different length scales

- It is secretly a **quantum circuit**

“entanglement at
different length scales”

and an **RG transformation**

“removes
short-range
entanglement”

$$|\Psi\rangle \rightarrow |\Psi'\rangle$$

$$H \rightarrow H'$$

“preservation
of locality”

- Entanglement entropy and correlations as in 1+1 critical ground states

$$S_L \approx \log(L) \quad C(L) \approx L^{-2\Delta}$$

blah, blah, blah... However, does it work?

[Given lattice Hamiltonian H ,
optimize variational parameters by energy minimization]

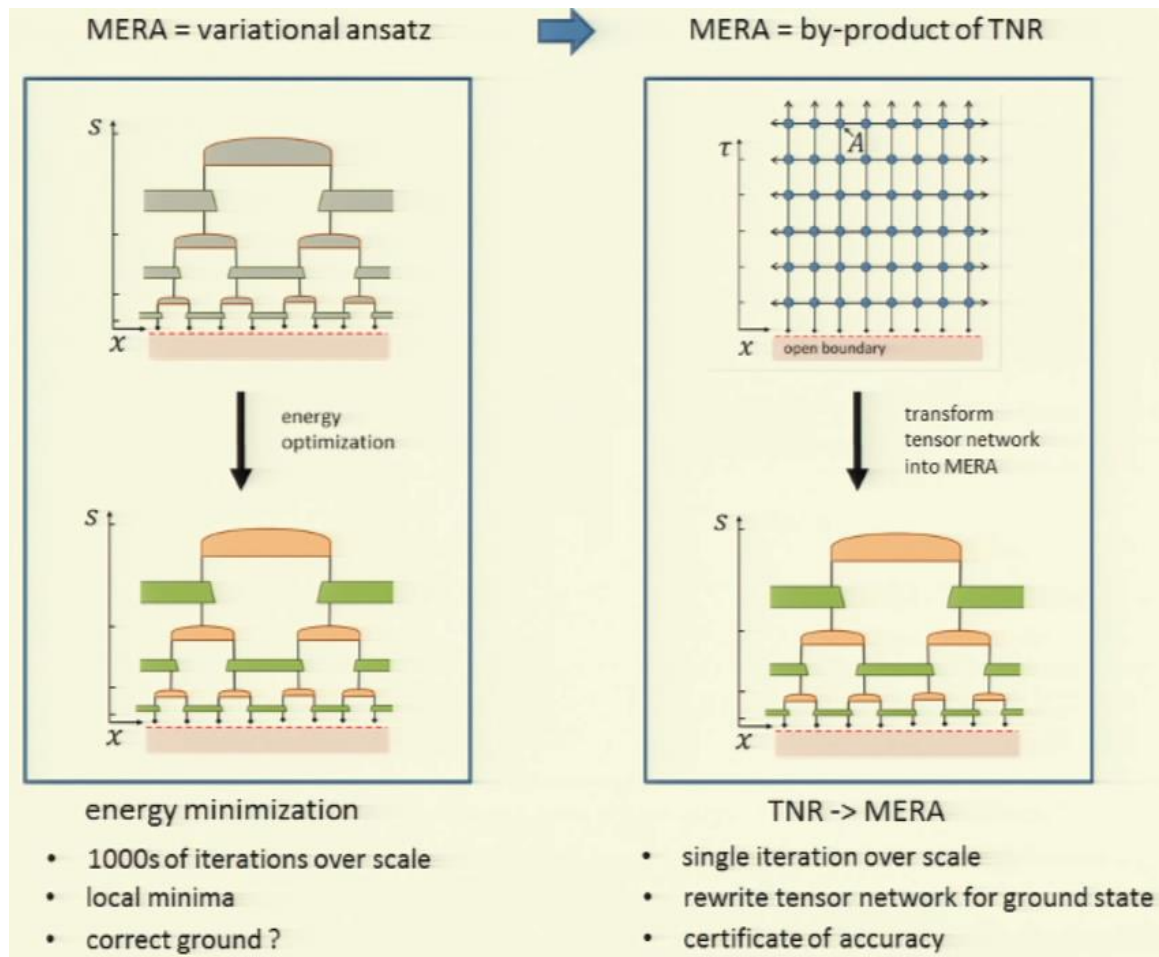
Critical Quantum Ising Models



Scaling dimensions of primary fields

		scaling dimension (exact)	scaling dimension (MERA)	error
identity	$\mathbb{I}$	0	0	----
spin	σ	0.125	0.124997	0.003%
energy density	ε	1	0.99993	0.007%
disorder	μ	0.125	0.1250002	0.0002%
fermions	ψ	0.5	0.5	$<10^{-8}\%$
	$\bar{\psi}$	0.5	0.5	$<10^{-8}\%$

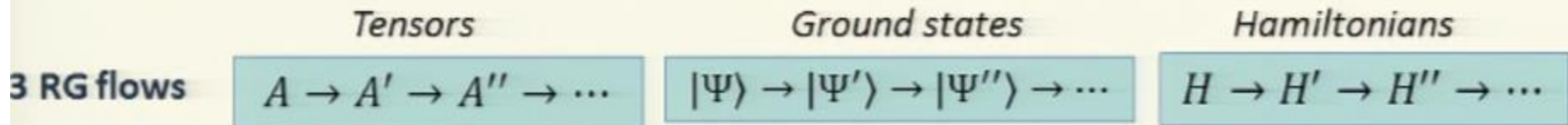
Ising Models



Ising Models

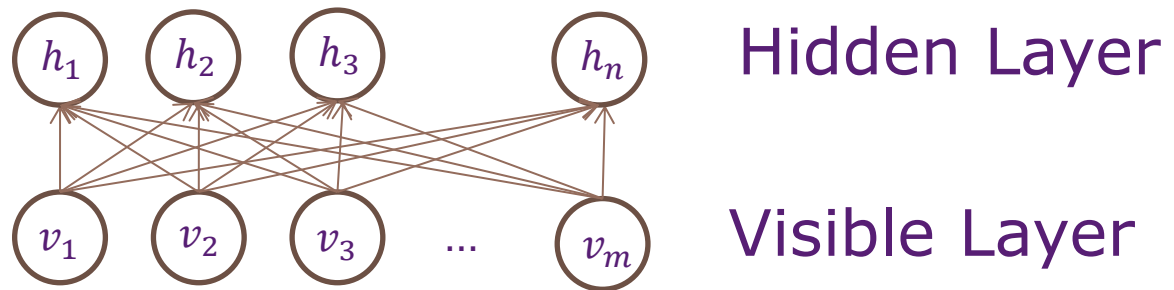


- Reformulation of the RG using quantum information tools/concepts (quantum circuits, entanglement)
- Efficient representation of ground states (MERA)
-> toy model for holography
- universal, non-perturbative, real-space RG approach



- Key ingredient: **removal of short-range entanglement**
- Very accurate in 1+1 dimensions (Ising model, etc)

RBM : Restricted Boltzmann Machine



$$P(\theta|v, h) = P(v_1) \dots P(v_n) P(h_1|v_1) \dots P(h_n|v_n) \dots P(h_n|v_1) \dots P(h_n|v_n)$$

$$P(\theta|v, h) \sim e^{\sum_{i=1}^n \sum_{j=1}^m w_{i,j} v_i h_j + \sum_{i=1}^n b_i v_i + \sum_{j=1}^m d_j h_j}$$

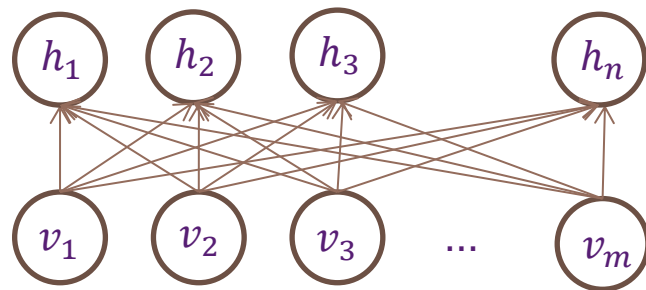
Interaction Energy Input Biases Output Biases

$$P(\theta|v, h) = \frac{e^{-E(v, h)}}{Z}$$

Energy

Partition Function

RBM : Restricted Boltzmann Machine (2)



Hidden Layer

Visible Layer

$$P(\theta|v, h) = P(v_1) \dots P(v_n) P(h_1|v_1) \dots P(h_n|v_n) \dots P(h_n|v_1) \dots P(h_n|v_n)$$

$$P(v|h) = \prod_{i=1}^n P(v_i|h) \quad P(h|v) = \prod_{i=1}^n P(h_i|v) \quad P(h_i = 1|v) = \sigma \left(b_i + \sum_{j=1}^m w_{i,j} v_j \right)$$

$$P(v_i = 1|h) = \frac{\exp(a_i^k + \sum_{j=1}^m w_{i,j}^k h_j)}{\sum_{p=1}^K \exp(a_i^p + \sum_{j=1}^m w_{i,j}^p h_j)}$$

Activation function
 $\sigma(x) \in [0,1]$

Normalization

Finite set of possible value for v



Renormalization for real neurons networks

Neurons are extended objects
Interactions are not (very) local

**real space = coarse-grain
by grouping with neighbors**

**coarse-grain by grouping with
most correlated partners**

**momentum space = coarse-grain
by filtering high frequencies**

**coarse-grain by filtering
eigenmodes of covariance matrix**

Let's look at some data.

**Caution: everything which follows is phenomenology.
And we should be cautious (!).**

Gaussian DBN : the Energy

$$E[v, h, \cdot | \theta] = \sum_{i=1}^{N_V} \frac{(v_i^E - b_i^E)^2}{2\sigma_i^2} - \sum_{i=1}^{N_V} \sum_{j=1}^{M_1} \frac{v_i^E}{\sigma_i^2} h_j^1 w_{ij}^E - \sum_{i=1}^{N_b} v_i^b b_i^b - \sum_{i=1}^{N_b} \sum_{j=1}^{M_1} v_i^b h_j^1 w_{ij}^b - \sum_{l=1}^{N_L} \sum_{j=1}^{M_1} h_j^l b_j^l - \sum_{l=1}^{N_L-1} \sum_{j=1}^{M_L} \sum_{k=1}^{M_{L+1}} h_j^l h_k^{l+1} w_{jk}^l - \epsilon \sum_{j=1}^{M_1} h_j^1$$

$$p[v_i^E | h^1, \theta] = \mathcal{N}\left[v_i^E, \sum_{j=1}^{M_1} h_j^1 w_{ij}^1 + b_i^E, \sigma_i\right] \quad \text{where } \mathcal{N}[x, \mu, \sigma] = \frac{e^{-\frac{(x-\mu)^2}{2\sigma^2}}}{\sigma \sqrt{2\pi}}$$

$$E[v_i^b | h, \theta] = - \left(b_i^b + \sum_{j=1}^{M_1} h_j^1 w_{ij}^b \right) v_i^b$$

$$p[v_i^b = 1 | v^E, h, \theta] = f\left[b_i^b + \sum_{j=1}^{M_1} h_j^1 w_{ij}^b\right] \quad \text{where } f[x] = \frac{1}{1 + e^{-x}}$$

$$p[h^1 = 1 | v, h^2, \theta] = f\left[\sum_{i=1}^{N_b} \frac{v_i^E}{\sigma_i^2} w_{ij}^E + \sum_{i=1}^{N_b} v_i^b w_{ij}^b + \sum_{k=1}^{M_2} h_k^2 w_{jk}^1 + b_j^1\right] \quad \text{where } f[x] = \frac{1}{1 + e^{-x}}$$

« mean-Field Approximation » for RBM

$$E[v, h|\theta] = -\sum_{i=1}^N v_i a_i - \sum_{i=1}^N \sum_{j=1}^M v_i h_j^{n-1} w_{i,j} - \sum_{j=1}^N h_j^{n-1} b_j$$

$$E[v|\theta] = -\text{Log}(\sum_{j=1}^N e^{-E[v, h_j^{n-1}|\theta]}) \quad \text{Energy } v$$

We define h^n to match $E[v|\theta]$ naturally so h_j^n by h_j^n

$$\begin{aligned} \text{Log}\left(\sum_{j=1}^N e^{\sum_{i=1}^N \sum_{j=1}^M v_i h_j^{n-1} w_{i,j} + \sum_{j=1}^N h_j b_j}\right) &= \sum_{i=1}^N \sum_{j=1}^M v_i h_j^n w_{i,j} + \sum_{j=1}^N h_j^n b_j \\ \Rightarrow -\sum_{i=1}^N \sum_{j=1}^M v_i h_j^{n-1} w_{i,j} - \sum_{j=1}^N h_j^{n-1} b_j &= -\sum_{i=1}^N \sum_{j=1}^M v_i h_j^n w_{i,j} - \sum_{j=1}^N h_j^n b_j \end{aligned}$$

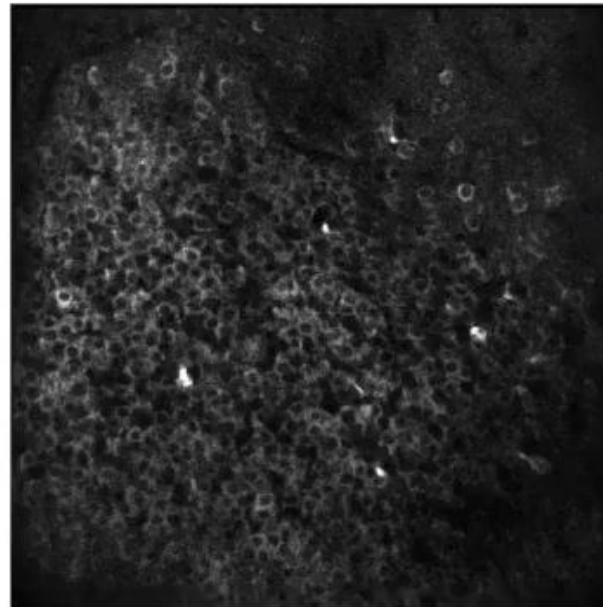
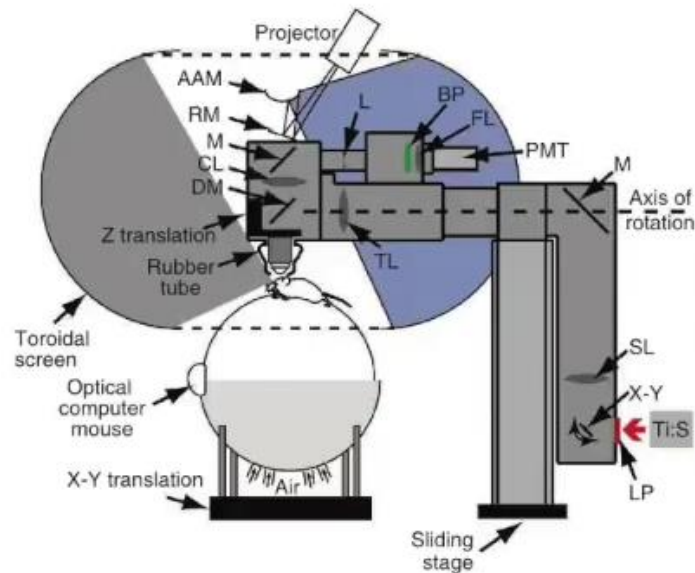
Relationships with the real brain



Observing Real neurons

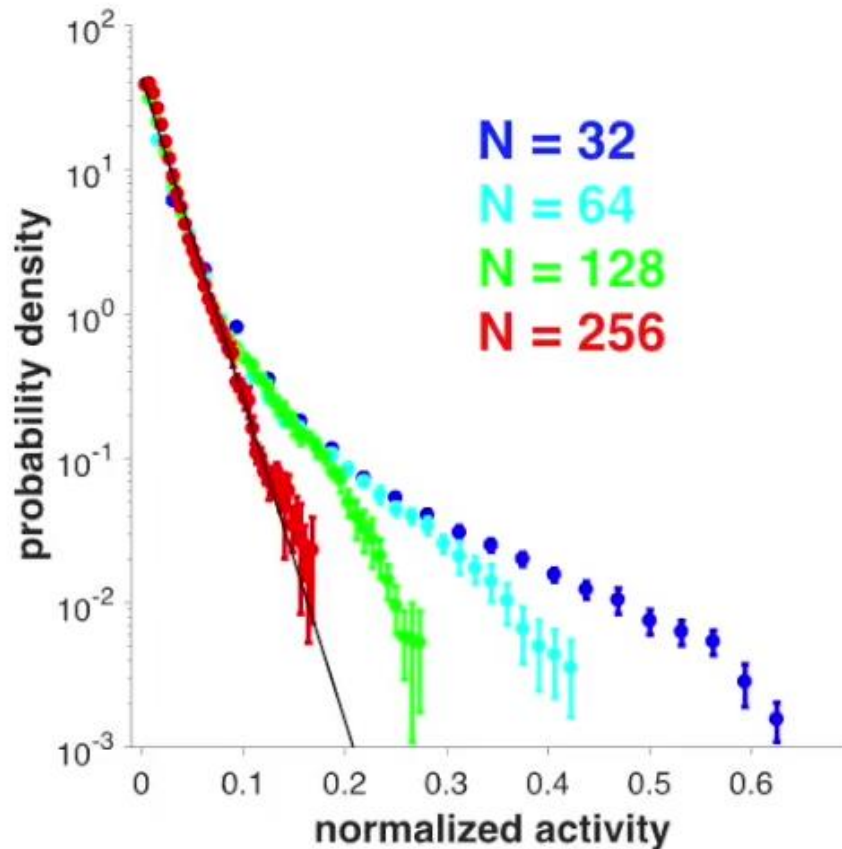


Optical recording from hippocampal neurons as a mouse moves in a virtual environment



DA Dombeck, CD Harvey, L Tian, LL Looger, and DW Tank, *Nat Neurosci* 13:1433 (2010).
L Meshulam, JL Gauthier, CD Brody, DW Tank, and WB, *Neuron* 96:1 (2017).

Distribution of Activity



**Distribution
approaches a fixed
form at large scales.**

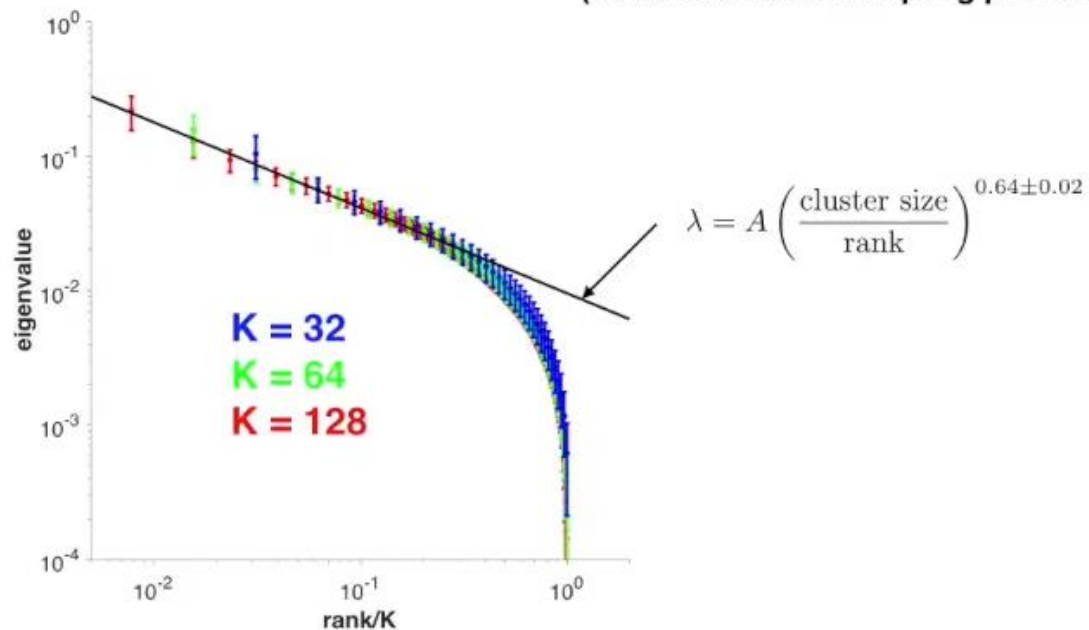
**This is also visible in
raw fluorescence data.**

Correlations Inside the Clusters



$C_{ij} = \langle \sigma_i \sigma_j \rangle - \langle \sigma_i \rangle \langle \sigma_j \rangle$

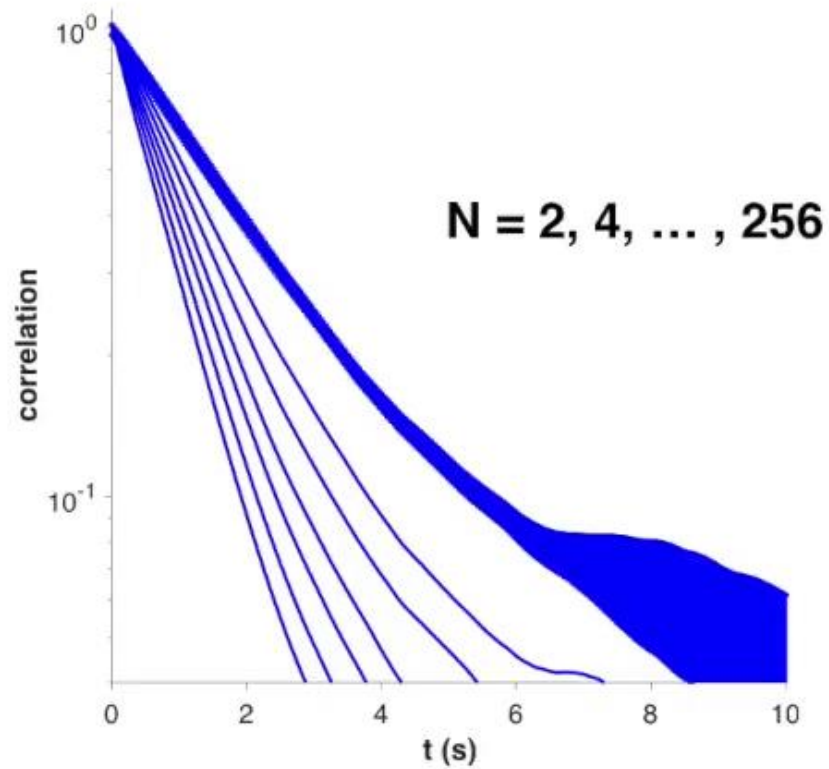
Find the eigenvalues in clusters of different sizes
(be careful about sampling problems!)



Dynamics of the Clusters



Larger clusters have slower dynamics ...



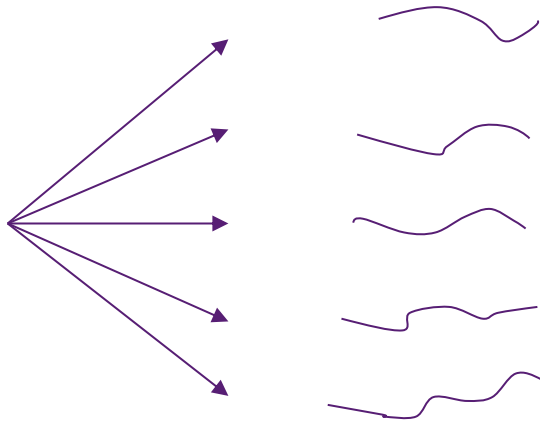
Application to Finance



Fast Pricing of Exotics (I)



100K-500K trajet.	1 – 10 years daily	1 – 100 underlyings	Realistic Models	Multiple features
-------------------	--------------------	---------------------	------------------	-------------------



Index
Dow 30
S&P 500
Nasdaq
SmallCap 2000
S&P 500 VIX
S&P/TSX
Bovespa
S&P/BMV IPC
DAX
FTSE 100
CAC 40
Euro Stoxx 50
AEX
IBEX 35
FTSE MIB
SMI

Local Volatilities

Stochastic
Interest rates

Stochastic FX

Correlation matrix

Capped
Floored
Asian
Strikes
Callable
Resets
Memory
PDI
Barriers
....

10 mins

Needed pricings: 100 prices :

Sensitivities
Exposures
Value at risk
Hedge Ratios



Fast Pricing of Exotics (II)



Nightly



1 M prix



Neural Network

1

Prices computation

2

Learning

23 parameters

	M	N	O	P	Q	R	S	T	U	V
bonus	YetiBarrier	YetiCoupon	PhoenixBarrier	PhoenixCoupon	PDBarrier	PDIGearing	PDIStrike	PDIType	maturity	
168	1,598662045	102,7934506	1,803550383	89,10621421	1,503284327	61,35308533	-1,131986778	61,16504473	-0,981872169	2,1633824
748	-0,865687113	102,5289398	1,207499897	93,12703428	0,629783047	62,28204399	1,906473794	59,70435676	0,206159686	4,2224176
888	0,511093603	90,91612936	1,721972548	94,80305415	1,92341021	49,31570746	-2,264827223	59,15778854	0,726806872	2,6569599
632	1,873411586	105,4550804	1,580305804	90,82321253	1,016549817	41,40652	2,741327755	67,40190344	0,587935765	4,3467833
925	-0,024438552	102,3634059	1,858138975	97,85346983	0,802054037	43,9653955	1,853283181	51,66273998	0,489263864	2,8627828
774	0,968667823	106,7197266	0,120575059	91,6186545	0,509863802	65,75974463	-1,767057384	61,62644528	0,463113878	1,6978381
827	-1,859266393	101,6584001	1,15385071	88,00817664	1,251674194	53,84697645	4,660926601	43,58051936	0,952898331	1,2988642
558	1,38549049	100,2770735	0,874996886	90,33014487	1,088720604	68,02265178	-0,962557176	57,20784498	0,011707592	2,9695811
014	-1,609906608	98,63363177	0,459939613	88,59774229	1,681616988	48,9917834	4,847473595	69,79549342	-0,506913407	3,0123358
902	0,212937624	94,98203874	0,143701045	91,98319076	1,063787317	60,32314953	-1,005645658	63,8324717	0,530987777	3,8815812
393	0,286776665	105,1712089	0,628513979	83,83704042	1,252492095	55,76686052	-0,286061138	53,32160594	-0,304853521	2,1418366
555	1,826821038	97,03536388	0,930870132	91,3584631	0,94559383	47,00123052	-2,243829941	56,50012468	0,000312713	2,2628155
117	1,6488869155	109,9430667	0,818865586	90,69036591	0,984073955	51,52967911	4,705981603	41,04753564	-0,091951987	1,2464048
668	1,797335297	91,58462697	1,321602856	86,86102174	1,494815143	67,82762906	-4,088887842	67,759794	-0,284710253	4,2824016

price

W	X	Y
ates	price	forw
13	28,15544	73,7
7	3,711043	67,8
19	33,21414	80,1
6	14,59038	77,9
18	16,84421	88,2
11	7,513129	83,0
20	-0,06915	93,6
8	14,32587	73,3
10	-2,54621	72,5
7	1,779381	75,4
12	7,629561	76,3
15	21,07478	84,8
21	11,57243	91,7
20	22,14178	79,7
5	-0,19858	111,

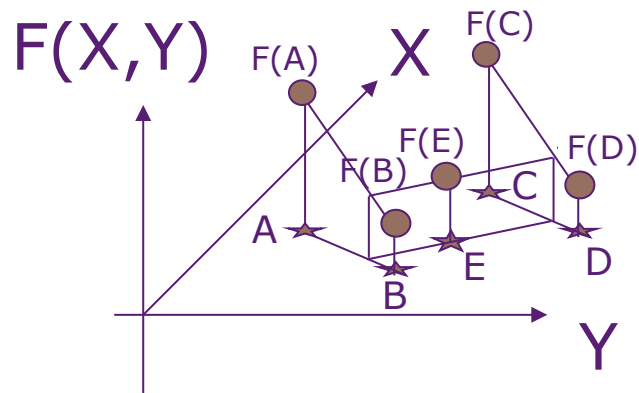
Name of the Game

Non-Linear Interpolation in 23 Parameters

Fast Pricing of Exotics (III)



Linear Interpolation in 2 Dimensions



Linear Interpolation in n Dimensions



2^n Surrounding points

Multilevel Sparse Gaussian
Kernels for interpolation

$N \leq 10$

1 M points



Less than 2 pts / dimension

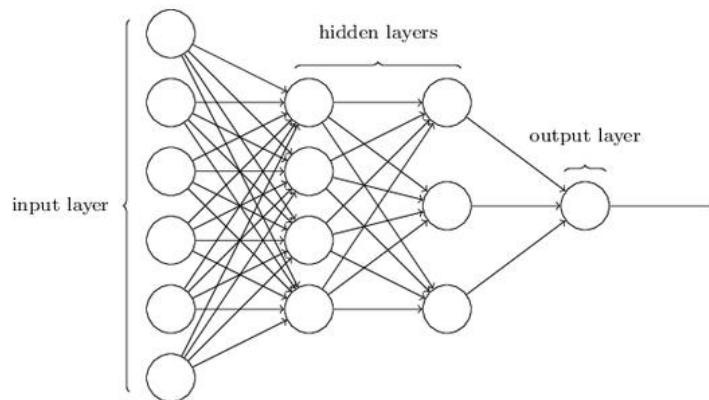
Fast Pricing of Exotics (IV)



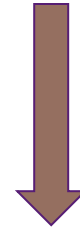
Back Propagation of Errors



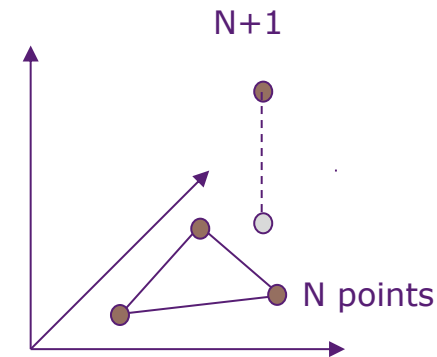
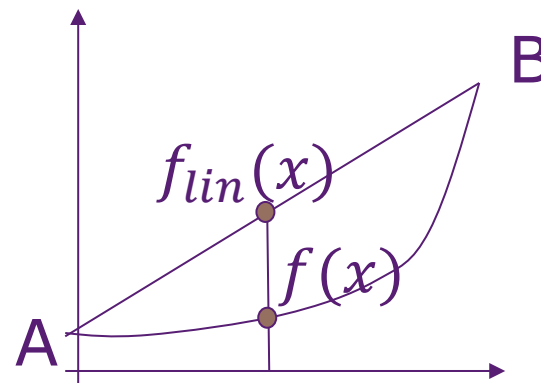
Kernel-like Effect



Deep network



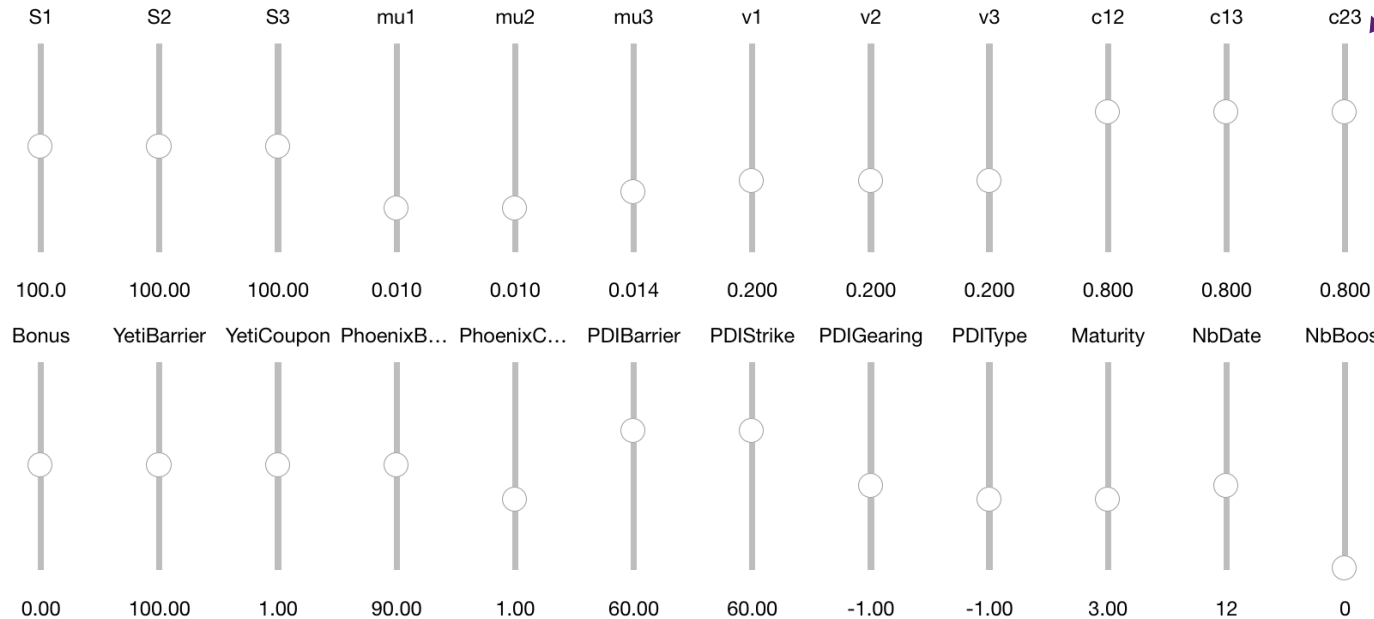
Non-Linearity



Fast Pricing of Exotics (V) Demo



```
1 PricingInterface(recompute_interface2, True).display()
```



Parameters

Present Price = 2.6134 Delta S1 = -0.0438 Delta S2 = -0.0438 Delta S3 = -0.0438
 Vega S1 = -4.3834 Vega S2 = -4.3834 Vega S3 = -4.3834

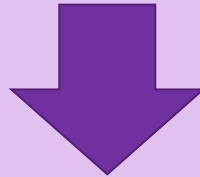
Present prices and 6 sensitivities



Next:

From Black Scholes world to Local Vol World

Introduce volatility surfaces, term structures



Go from 23 dimensions -> 500 dimensions

Smart Optimization





Why it should work

Why do the simplex algorithm work in practice ?

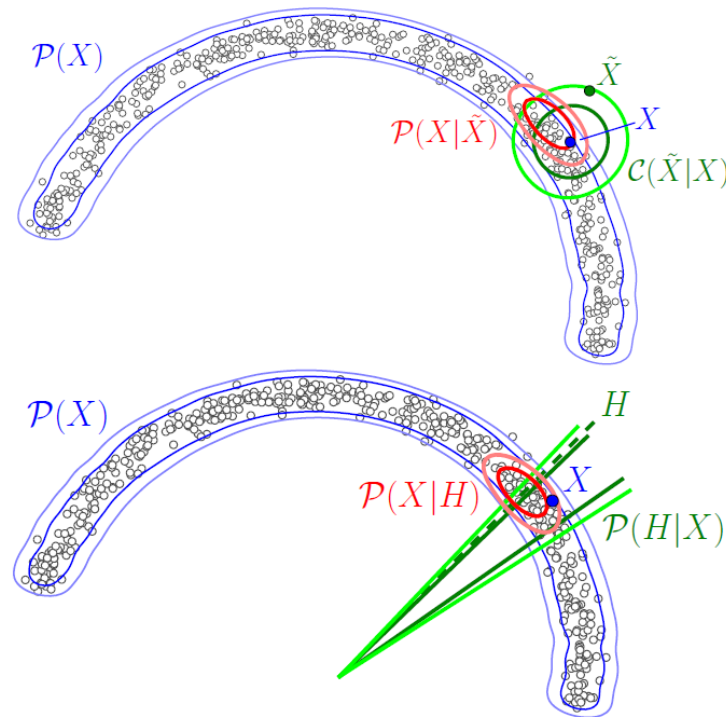
Manifold Tangent Classifier for mouvement understanding

We will build a Manifold tangent pricer

Manifold Tangent Hypothesis : Example



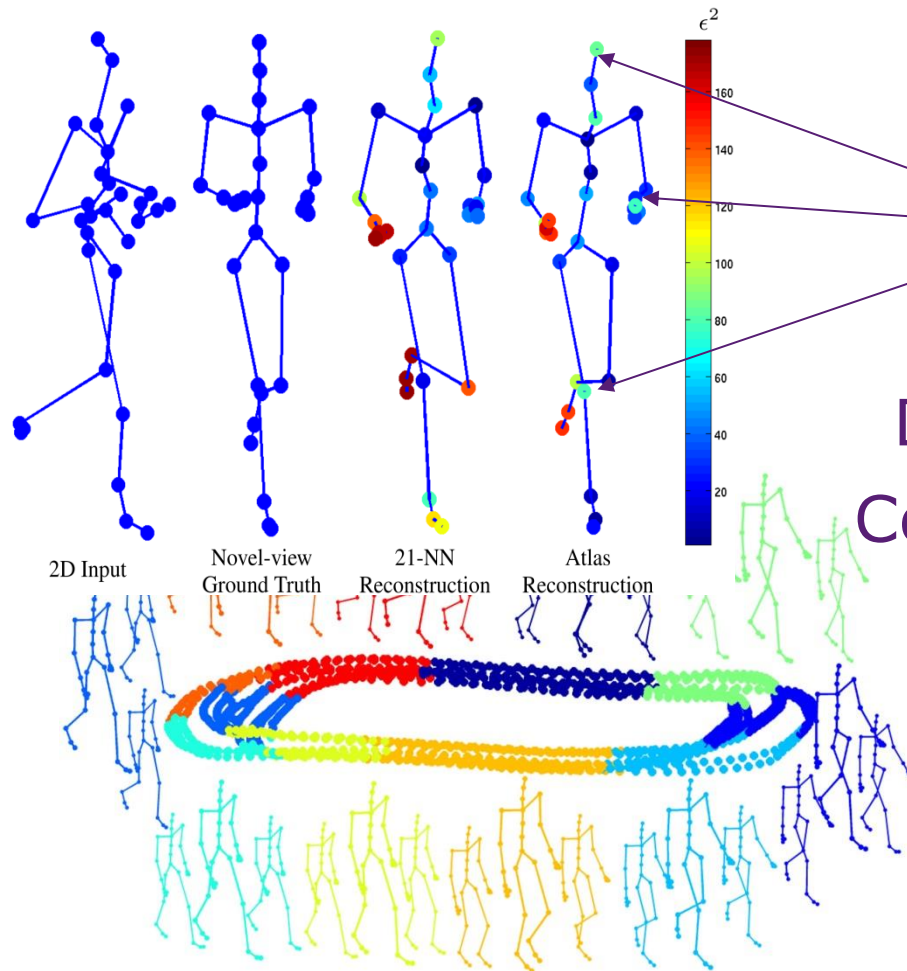
While the data is presented in a high dimensional space, locally, the data is organized in a small dimension effective space.



<http://colah.github.io/posts/2014-03-NN-Manifolds-Topology/>

Salah Rifai, Yann N. Dauphin, Pascal Vincent, Yoshua Bengio, Xavier Muller

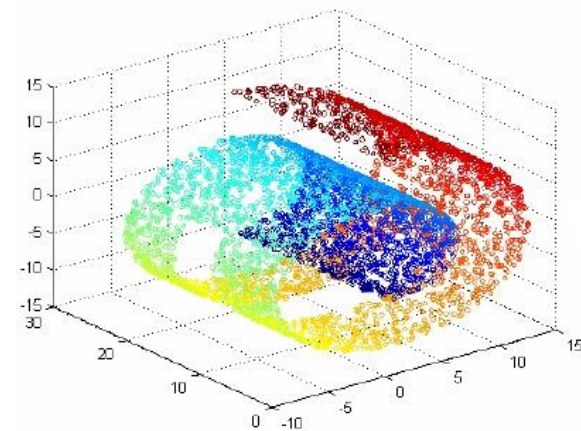
Manifold Tangent Hypothesis : Example



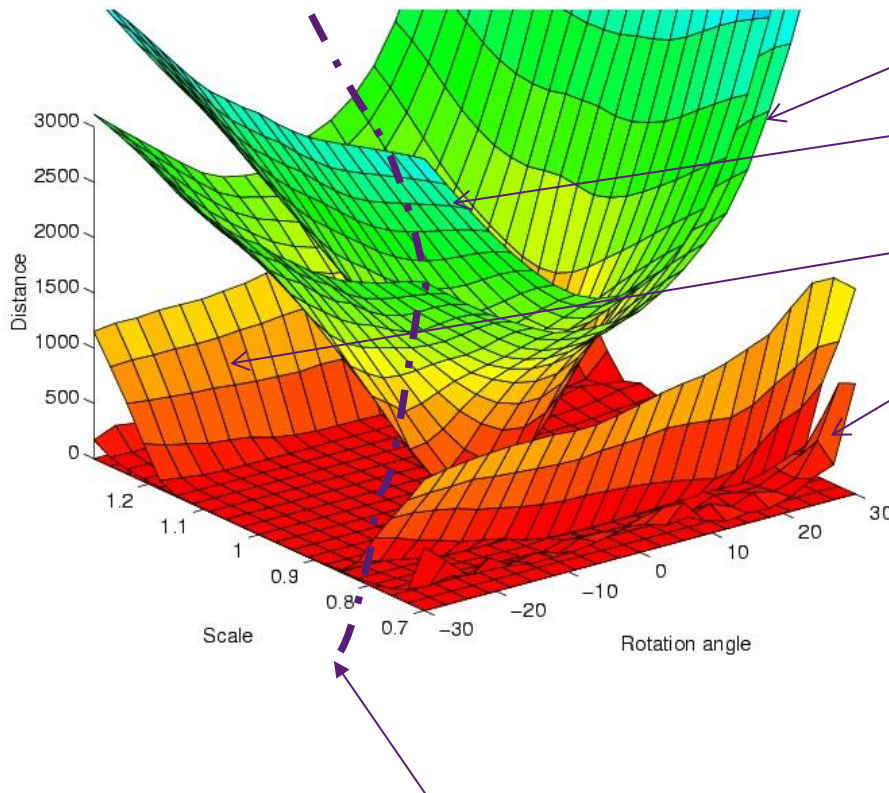
A lot of sensors s

Data = $\{x_s(t), y_s(t), z(t)\}$

Constraints : articulations



Isoprice Surfaces



Classe price = 1

Classe price = 2

Classe price = 3

Classe price = 4

Constraints:
Non-Arbitrage,
Consistency,..

Transverse price sub-space (line here)

